

Watch What They Do, Not What They Say: Estimating Regulatory Costs from Revealed Preferences

Adrien Alvero

Columbia Business School, USA

Sakai Ando

International Monetary Fund, USA

Kairong Xiao

Columbia Business School, USA

We show that distortion in the size distribution of banks around regulatory thresholds can be used to identify costs of bank regulation. We build a structural model in which banks can strategically bunch their assets below regulatory thresholds to avoid regulations. The resultant distortion in the size distribution of banks reveals the magnitude of regulatory costs. Using U.S. bank data, we estimate the regulatory costs imposed by the Dodd-Frank Act. Although the estimated regulatory costs are substantial, they are significantly lower than banks' self-reported estimates. (*JEL* G21, G28)

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Many regulatory agencies are required—by statute or executive order—to perform a cost-benefit analysis (CBA) on a proposed rule.¹ A CBA on financial

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¹ The current set of requirements includes Executive Order 12866 and Office of Management and Budget (OMB) Circular A-4, the Regulatory Flexibility Act (RFA), and the Unfunded Mandates Reform Act (UMRA). These

regulation usually compares the social benefits of regulations, such as those arising from a reduced probability of a financial crisis, with the regulatory costs, which are mainly borne by financial institutions that must comply with them. A CBA could form the basis for judicial review and Congressional oversight of regulatory actions.² Despite a growing body of research on quantifying regulatory benefits to the public, quantifying regulatory costs borne by financial institutions is often viewed as a mundane task interesting only to regulators, and as a result, it has received less academic scrutiny.

Quantifying the costs of financial regulation, however, is far from trivial. Financial regulators generally do not have enough information to gauge the regulatory impacts on complex financial institutions. Instead, they often have to rely on self-reported estimates from financial institutions to guide their policies. However, these self-reported estimates may be inflated by financial institutions to justify regulatory relief. For instance, Parker (2018b,a) suggests that a highly influential estimate of regulatory costs cited by the 2016 House Concurrent Budget Resolution comes from studies “funded by organizations having a strong financial or institutional stake in the outcome of their studies” and the methodologies used to estimate regulatory costs are “fundamentally flawed.” Nevertheless, these self-reported estimates are widely cited in policy debates and have supported efforts to roll back financial regulations enacted after the 2008 financial crisis.³ The lack of independent and rigorous assessment of regulatory costs highlights the need for academic research in this area (Posner and Weyl 2013; Coates 2014).

This paper proposes a revealed preference approach to infer regulatory costs from the regulatory distortion on the size distribution of financial institutions. As an illustration of our approach, consider the Dodd-Frank Act of 2010, which imposes additional regulations on banks when their assets cross the \$10 billion and \$50 billion thresholds.⁴ As shown in Figure 1, panels A and B, banks shrink their assets to avoid these regulations, creating excess densities around these thresholds. We show that the extent of such regulatory distortions can reveal the regulatory costs faced by banks.

To formalize this idea, we develop a structural model of the size distribution of banks. In the model, regulatory costs are modeled as a tax on banks’ profits

requirements vary in terms of the agencies and rules they cover and the types of analyses required (Carey 2014). Independent financial regulators are exempt from Executive Order 12866 requirements but are subject to other requirements to assess the effects of their rules. See Congressional Research Service (CRS) Report R42821, *Independent Regulatory Agencies, Cost-Benefit Analysis, and Presidential Review of Regulations*.

² Judicial review is a right imbued by statute, but some elements of analysis can be shielded from judicial review.

³ For instance, in January 2017, the Trump administration issued Executive Order 13771, which mandates “for every one new regulation issued, at least two prior regulations be identified for elimination” (this order was revoked by Executive Order 13992 in January 2021). In May 2018, Congress passed the Economic Growth, Regulatory Relief, and Consumer Protection Act, which scales back many financial regulations enacted after the crisis.

⁴ We provide a more detailed discussion on the regulations triggered by each threshold in Section 1.

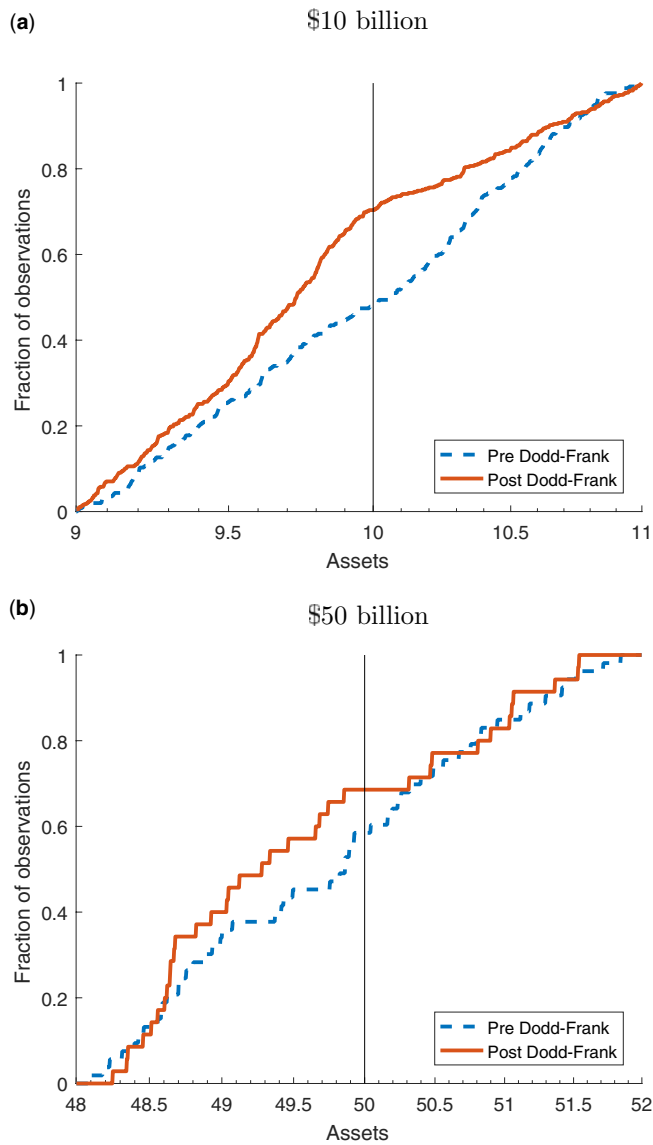


Figure 1
The size distribution of banks around regulatory thresholds before and after Dodd–Frank
This figure shows the empirical cumulative distribution of bank size conditional on the narrow intervals around the \$10 billion and \$50 billion regulatory thresholds. Bank size is measured by quarter-end assets. The blue dashed line and the red solid line represent the pre-Dodd-Frank period (2001Q1 to 2007Q3) and post-Dodd-Frank period (2010Q3 to 2018Q2), respectively.

in the spirit of Posner (1971). The tax rate jumps discretely at regulatory thresholds, reflecting the regulations that come into effect once banks' assets exceed these regulatory thresholds.⁵ The discrete jumps in the tax rate create an incentive to bunch. Banks that are just above the regulatory threshold shrink to avoid regulation. Banks that are far above the regulatory threshold do not bunch because the costs of staying below the threshold outweigh the costs of regulation.

We use the model to guide our estimation of the regulatory costs. In the absence of regulatory distortions, the size distribution of banks should not display any excess density around a particular threshold. In the presence of regulation, however, banks bunch to avoid regulation, creating excess densities around regulatory thresholds. The magnitude of the excess densities then reveals the regulatory costs borne by banks. Using this idea, we derive a maximum likelihood estimator and apply it to U.S. bank size data to estimate the regulatory costs imposed by the Dodd-Frank Act. The key identification assumption is that the bank size distribution does not display the excess densities around the size cutoffs shown in Figures 1a and 1b in the absence of the Dodd-Frank Act. This assumption is consistent with the fact that the distortions around the size thresholds are not present in the pre-Dodd-Frank period or around other round numbers that are not regulatory thresholds, such as \$20 billion or \$40 billion, as shown in Figure 2, panels A and B.

We estimate the model using bank size data in the post-Dodd-Frank period. The estimation shows that the regulation costs triggered at the \$10 billion threshold are equivalent to a 0.41% tax on banks' average annual profits. The additional regulatory costs triggered at the \$50 billion threshold are equivalent to a 0.11% tax. The estimated regulatory costs are economically significant. For a bank with \$50 billion assets, the regulatory costs amount to \$4.16 million per year,⁶ which is equivalent to the salary expenses of hiring 52 additional compliance officers, assuming the average annual compensation for a compliance officer is around \$80,000 (Feldman, Heinecke, and Schmidt 2013).

Regulation not only imposes compliance costs for banks but also leads to indirect costs for the rest of the economy. As banks shrink their size to avoid regulation, bank-dependent firms could be adversely affected if banks shrink lending.⁷ Regulation could also affect credit supply from the extensive margin

⁵ This formulation also allows for the possibility that certain regulations may generate private value for banks that comply with them. In this case, the tax rate should be interpreted as the net regulatory burden. See Section 2.1 for a more detailed discussion.

⁶ The regulatory costs introduced by the Dodd-Frank Act for a \$50 billion bank are equivalent to $0.41\% + 0.11\% = 0.52\%$. We sum the regulatory tax triggered at \$10 billion and \$50 billion to calculate the total Dodd-Frank burden for a \$50 billion bank because it is subject to the regulations triggered at both \$50 billion and \$10 billion. We then calculate the dollar value of regulatory costs per year by multiplying the regulatory tax rate with the average profits, $\$50,000M \times 1.6\% \times 0.52\% = \$4.16M$, where 1.6% is the ratio of average annual profits to assets.

⁷ See Bowman, Hu, and Johnson (2018) for evidence that banks near the Dodd-Frank threshold decrease their lending.

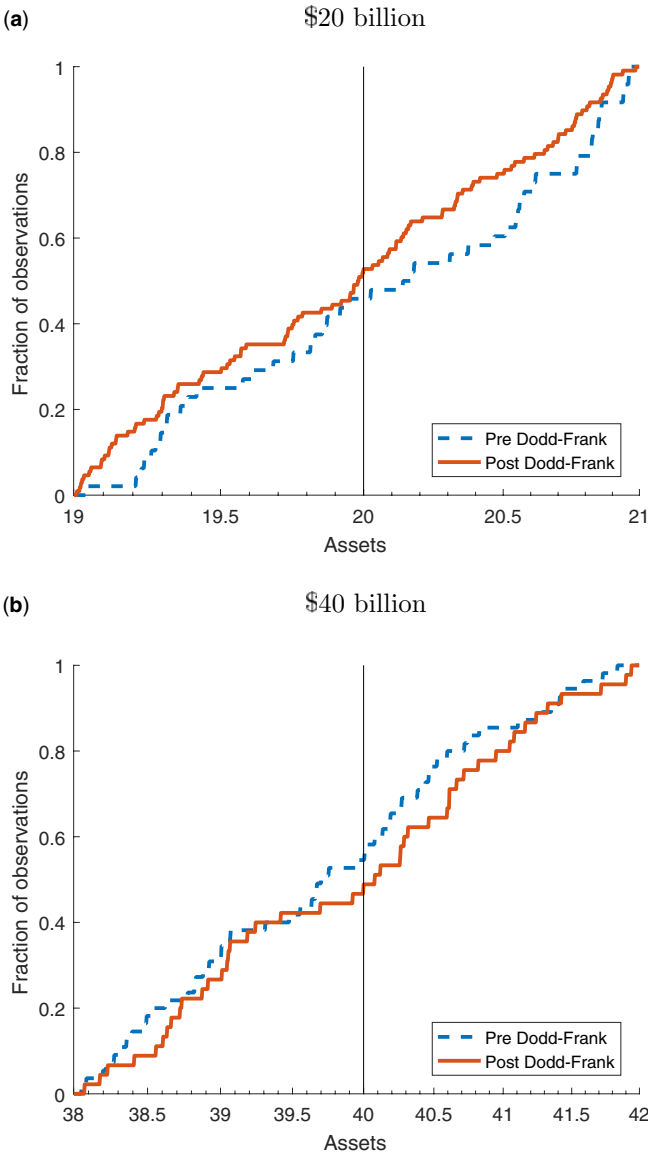


Figure 2
The size distribution of banks around nonregulatory thresholds before and after Dodd-Frank
This figure shows the empirical cumulative distribution of bank size conditional on the narrow intervals around \$20 billion and \$40 billion. Bank size is measured by quarter-end assets. The blue dashed line and the red solid line represent the pre-Dodd-Frank period (2001Q1 to 2007Q3) and post-Dodd-Frank period (2010Q3 to 2018Q2), respectively.

of bank entry and exit.⁸ Quantifying the indirect costs of regulation requires solving the full market equilibrium. To this end, we calibrate parameters that govern the equilibrium supply and demand of bank credit using either values from the prior literature or the corresponding moments in the data. We then simulate counterfactual economies with and without the direct regulatory costs estimated from our maximum likelihood estimator. We quantify the indirect costs of regulation as the lost output of bank-dependent firms due to the regulation. We find that the indirect costs of Dodd-Frank Act appear to be modest: the indirect regulatory costs are equivalent to a 0.02% tax on the output of bank-dependent firms.

It is worth noting that estimating the indirect costs requires stronger assumptions and more parameters on the production functions of banks and firms. First, we assume that banks have the same business model and lend to the same representative firm. This assumption simplifies the model but prevents us from studying the heterogeneity of indirect costs across firms. Second, we assume that potential entrants can choose to be a bank or not to enter the lending market at all. In reality, entrants can also choose to be a shadow bank, which could affect the quantification of indirect costs. Finally, the model focuses on the industrial equilibrium of the banking sector. The estimates should not be interpreted as the impact of regulation on the aggregate credit supply because firms can substitute loans with other forms of credit, such as issuing corporate bonds. The regulatory impacts may also spill over to households through labor and product markets. Nevertheless, the small effects in partial equilibrium are interesting on their own, as they are not in line with the typical banker view.

Practitioners often point to the low franchise values in the postcrisis period as evidence of high regulatory costs imposed by the Dodd-Frank Act. Although this argument is appealing, it remains unclear how much regulations can explain the decline in bank values (Sarin and Summers 2016; Chousakos and Gorton 2017). Our structural model can study this question quantitatively by simulating bank values with and without the estimated regulatory costs. We find the estimated regulatory costs only explain a small fraction of the decrease in bank franchise value in the postcrisis period. Our result suggests nonregulatory factors, such as an ultra-low-interest-rate environment (Calomiris and Nissim 2014; Whited, Wu, and Xiao 2021), market reassessment of risks (Sarin and Summers 2016), and the removal of too-big-to-fail subsidies (Atkeson, d'Avernas, Eisfeldt, and Weill 2019; Berndt, Duffie, and Zhu 2020), may also have contributed to the decline in bank values.

The Dodd-Frank Act also has distributional implications in the cross-section of banks. One may expect the market shares of big banks (>\$50 billion) would shrink after the Dodd-Frank Act because they face the most stringent regulations. Interestingly, the counterfactual simulation suggests that

⁸ It is possible that firms can substitute to shadow banks, so firms' overall credit access is not affected. We discuss this issue in Section 3.4.

the market share of big banks would expand after the Dodd-Frank Act. There are two reasons for the expansion of big banks. First, medium-sized banks (\$10 billion to \$50 billion) engage in regulatory avoidance, which reduces their average asset size. Second, heightened regulatory costs depress bank value across the size spectrum, which reduces the entry of small banks (<\$10 billion). This result suggests a possible unintended consequence of regulation on the bank industry's market structure.⁹

We compare our estimated regulatory costs with those from other methodologies. We show that reduced-form methods, such as difference-in-differences and regression-discontinuity design, likely underestimate the direct regulatory costs because banks can strategically avoid regulation. Furthermore, these methods cannot capture implicit regulatory costs that are not measured in banks' financial statements. Indeed, these reduced-form methods usually find little evidence of changes in regulatory costs after the Dodd-Frank Act. In contrast, estimates based on self-reported surveys from banks are usually much larger than our estimates, consistent with anecdotal evidence that these estimates may be inflated to lobby regulators for regulatory relief (Hinkes-Jones 2017; Parker 2018a).

We conduct several extensions and robustness checks on our results. First, we extend our analysis to the 2018 Economic Growth, Regulatory Relief, and Consumer Protection Act of 2018, which rolled back many regulatory requirements imposed by the Dodd-Frank Act. In the data, the excess densities around the Dodd-Frank thresholds have decreased significantly after the 2018 regulatory relief, suggesting the incentive to avoid regulation has weakened since then. Indeed, the estimated regulatory costs in the postrelief period are significantly smaller than those in the post-Dodd-Frank but prerelief period. We also find that the number of banks in the model would increase after the 2018 regulatory relief, consistent with an increase in bank entry observed in the data.

Second, our maximum likelihood estimator assumes the undistorted bank assets follow a power-law distribution. Although this assumption is motivated by the data, we assess the robustness of our estimation to alternative distributions. To this end, we reestimate the regulatory costs using an alternative distribution assumption for the maximum likelihood estimator and find the results are robust. Our estimates are not sensitive to the distribution assumption because the regulatory costs are mainly identified from the local abnormal densities around the threshold and are insensitive to the global property of the distribution.

Third, our baseline estimation uses all the post-Dodd-Frank years to estimate the regulatory cost. The implicit assumption is that the local size distribution

⁹ This result does not imply that the Dodd-Frank Act has worsened the too-big-to-fail problem because we do not explicitly model how regulation affects big banks' risk-taking and their reliance on the implicit government guarantee.

around the regulatory threshold has finished adjusting to the regulatory changes before the start of the estimation sample. However, if there are convex adjustment costs, the bank size distribution may take time to adjust to the regulatory change. The still-in-progress changes in the size distribution may bias our estimates downward. To address this concern, we plot the bank size distribution around the regulatory thresholds year by year. We note that the size distribution adjusts to the regulatory changes quite fast, and the bunching pattern appears one year after the law's passage. We drop the first several years after the passage of the Dodd-Frank Act and rerun the estimation. We find that the estimated regulatory costs are quite similar to the baseline estimates, again consistent with the fact that bank size adjustment is relatively fast.

Finally, we conduct several placebo tests by applying our estimator to the pre-Dodd-Frank sample and find no regulatory costs triggered by the thresholds in this sample. We also conduct placebo tests on round numbers that are not regulatory thresholds in the post-Dodd-Frank period. Again, our estimator correctly indicates null results. We also show that our results are robust if we allow bank profit margins to be size-dependent or use alternative measures of bank regulatory assets.

Our paper relates to the literature on cost-benefit analysis (CBA) of regulation. An extensive literature investigates regulatory CBA in environmental economics and industrial organization (Harberger 1964; Viscusi and Aldy 2003). However, analogous literature has been conspicuously missing in financial economics until recently (Posner and Weyl 2013). The existing research focuses on quantifying the benefits of financial regulation.¹⁰ However, quantifying regulatory costs has received less attention so far. Our paper contributes a revealed preference approach to quantify regulatory costs.¹¹ This revealed preference approach is less prone to potential bias in self-reported surveys and complements reduced-form methods to estimate regulatory costs, such as difference-in-differences and regression discontinuity, for which endogenous selection around the threshold is an impediment for identification. Although our revealed preference approach addresses several limitations of the existing methods, it also has a few caveats. First, this approach needs to make assumptions about the data generating process. Robustness concerning these assumptions needs to be assessed. Second, this approach requires the observed behavior to incorporate the regulatory changes fully. One needs to run diagnostic tests to verify whether still-in-progress adjustments are present

¹⁰ Coates (2014) surveys the literature on the estimation of gross domestic product (GDP) losses due to financial crisis and how much bank regulation can reduce such losses. Posner and Weyl (2013) propose and estimate a parameter called the statistical cost of a crisis (SCC), which allows regulators to translate the reduced probability of a crisis into a dollar value with certainty.

¹¹ Some researchers have developed indexes of regulatory costs based on textual analysis of the regulatory provisions or compliance-related spending (Al-Ubaydli and McLaughlin 2017; Calomiris, Mamaysky, and Yang 2020; Simkovic and Zhang 2020). These indexes are helpful for understanding the cross-section variations in regulatory exposure, but they are difficult to translate into monetary values in a CBA.

in the data. Finally, similar to reduced-form methods, this approach provides an ex post assessment of regulatory costs based on observational data. It cannot replace the survey approach in assessing completely new regulations that have no observational data.

Our approach is related to the bunching literature in public finance and labor economics (Saez 2010; Chetty, Friedman, Olsen, and Pistaferri 2011; Kleven and Waseem 2013).¹² The classic bunching approach is to take a known discontinuity in a tax schedule and then estimate preference parameters, such as labor supply elasticity. In contrast, this paper takes banks' preferences as given and then backs out a tax equivalent of regulatory costs. In other words, our approach is a bunching estimator in reverse. In addition, our paper differs from the prior bunching studies by taking a full structural approach, which allows us to estimate the indirect regulatory cost and conduct a rich set of counterfactuals. By doing so, our paper adds to a growing body of literature that employs structural techniques to study financial markets (Kojen and Yogo 2015; Egan, Hortaçsu, and Matvos 2017; Buchak, Matvos, Piskorski, and Seru 2018; Benetton 2021; Nelson 2018; Robles-Garcia 2019; Darmouni 2020; Xiao 2020; Corbae and D'Erasmus 2021; Begenau and Landvoigt 2021).

This paper also relates to the vast literature on financial regulation. The aftermath of the 2008–2009 financial crisis witnessed one of the most active periods of financial regulation in U.S. history. The unprecedented wave of financial regulation has stimulated a fast-growing body of research. The existing literature shows from a variety of perspectives how the new regulations affect the financial system (Buchak, Matvos, Piskorski, and Seru 2018; Acharya, Berger, and Roman 2018; Cortés, Demyanyk, Li, Loutskina, and Strahan 2020; Fuster, Plosser, and Vickery 2018; Carletti, Goldstein, and Leonello 2020; Kashyap, Tsomocos, and Vardoulakis 2020). We contribute to this literature by quantifying the costs of Dodd-Frank regulations on banks. This question is crucial because the narrative of repressive regulatory costs has made many Dodd-Frank regulations targets for repeal. We find that the self-reported estimates from the finance industry are usually much larger than the estimates from the revealed preference approach, consistent with the anecdotal evidence that regulatory costs are often “hyped” by the finance industry (Hinkes-Jones 2017, p. 1). Furthermore, we study the impacts of the Dodd-Frank regulations on bank size, productivity, and profit distribution. By doing so, our paper contributes to the literature on the distortionary effects of size-based regulation, which has mainly focused on the labor market so far (Garicano, Lelarge, and Van Reenen 2016; Gourio and Roys 2014; Ando 2021).

¹² See also DeFusco and Paciorek (2017), Dagostino (2018), Bachas, Liu, and Yannelis (2019), Alvero and Xiao (2020), Bachas, Kim, and Yannelis (2020), and Antill (2020) for applications of the bunching approach in finance.

1. Institutional Background

We first discuss the cost-benefit analysis of financial regulation. Then we discuss the Dodd-Frank Act and its impact on the size distribution of banks.

1.1 Cost-benefit analysis of regulation

CBA entails an economic or statistical assessment of the social benefits of the regulation and the compliance costs borne by the regulated parties.¹³ The goal of CBA is to advance regulators' ability to increase welfare and allow the public to detect and push back against regulations that fail to do so. CBA often provides grounds for judicial review and Congressional oversight of regulatory actions.

CBA of financial regulations has emerged as an important point of policy debate since the passage of the Dodd-Frank Act. Proponents argue that financial regulators should be held accountable for quantified CBA to ensure their discretion on rulemaking is not abused (Posner and Weyl 2013). However, others caution about the inherent difficulties in conducting a CBA of financial regulations because of the complexity of financial institutions and the lack of reliable data (Coates 2014; Cochrane 2014).

1.2 The Dodd-Frank Act

The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 (Dodd-Frank) is the centerpiece of financial reforms initiated after the 2007–2008 financial crisis. The Dodd-Frank Act takes a tiered regulatory approach: banks are classified into size categories based on several thresholds. Larger banks are subject to stricter regulations. Specifically, banks whose assets exceed the \$10 billion threshold are required to (1) conduct annual stress tests, (2) comply with the Durbin Amendment, which puts a cap on the fees charged to merchants for debit card transactions, (3) report to the Consumer Financial Protection Bureau (CFPB), a government agency created as part of Dodd-Frank, and (4) create risk committees with independent directors. Banks whose assets exceed \$50 billion are subject to additional risk-based capital and liquidity requirements, stress tests, and annual resolution plans. A detailed summary of the law can be found in Huntington (2010).

The tiered system creates discontinuities in the burdens at the regulatory thresholds. In response to such discontinuities, banks around the thresholds strategically downsize their assets to avoid regulation. As represented by the red solid line in Figure 1, panels A and B, the cumulative distribution function of bank size displays abnormal bulges around the \$10 billion and \$50 billion thresholds after the passage of the Dodd-Frank Act, suggesting that banks bunch their assets to avoid regulation.¹⁴ This pattern is consistent

¹³ Appendix Section A.1 discusses the major statutes that require financial regulators to conduct CBA and a case study of a judicial review of a financial regulation based on the CBA.

¹⁴ Internet Appendix Figure OA.1 shows histograms of bank size around the regulatory thresholds.

with Bouwman, Hu, and Johnson (2018), who find that banks around the Dodd-Frank thresholds substantially reduce their assets.¹⁵ Such a pattern is not present in the pre-Dodd-Frank period, as represented by the blue dashed line in Figure 1, panels A and B.

One may worry that banks may simply cluster around round numbers and do not necessarily try to avoid regulation. To address this concern, we examine round numbers that are not regulatory thresholds, such as \$20 billion or \$40 billion. We find no excess density around these round numbers, as shown in Figure 2, panels A and B. This result suggests that the excess densities around \$10 billion and \$50 billion are not driven by banks' clustering at round numbers. Instead, they are the outcomes of banks' strategic response to the regulations imposed by the Dodd-Frank Act.

1.3 Bank size determination for regulatory purposes

The Dodd-Frank Act does not provide a uniform methodology to determine bank size for regulatory purposes. Instead, the implementation of different provisions was left to separate rulemaking processes, which led to slightly different methodologies to determine bank size. For instance, the annual stress test uses average assets in the past four quarters;¹⁶ the Durbin Amendment uses end of calendar year assets;¹⁷ and the CFPB requirement uses the minimum of the last four consecutive quarters with the exception that all institutions above \$10 billion as of June 30, 2011, would be required to report to the CFPB, and would only stop having to report if their total assets are below \$10 billion for four consecutive quarters.¹⁸ In this paper, we use the simplest measure of quarter-end assets to construct the size distribution and conduct different robustness checks with other methodologies to determine bank size. We will also allow a measurement error term in the estimation to account for these intricacies in the regulation.

2. Model

This section introduces our theoretical framework. We first describe individual banks' optimal size choices in the presence of the size-based regulation in a partial equilibrium setting in which the distribution of productivity and the lending rate are given. We derive a sufficient statistic formula for the direct regulatory costs. Then, we endogenize the productivity distribution and the lending rate in a general equilibrium framework with firms, which allows us to quantify the indirect costs of bank regulation on the rest of the economy.

¹⁵ This pattern is also consistent with Eisenbach, Lucca, and Townsend (2021), who find that regulators spent considerably more work hours on banks above \$10 billion in the postcrisis period.

¹⁶ See https://www.ffiec.gov/PDF/FFIEC_forms/FFIEC016_20171006_i_draft.pdf.

¹⁷ See <https://www.federalreserve.gov/paymentsystems/regii-faqs.html>.

¹⁸ See <https://www.fdic.gov/news/press-releases/2011/pr11179a.html>.

2.1 Bank size choice and direct cost of regulation

The economy is populated by heterogeneous banks indexed by their productivity z , following a probability density function $g(z)$. Although we refer to z as productivity, one should interpret z broadly as any nonregulatory factors that affect banks' preferred size. For example, the value of z might vary with banks' market power and deposit base, among other factors. Higher values of z correspond to larger potential bank size in the absence of regulatory distortion. A bank can raise funding from depositors at a cost $r(q|z)$, where q is the logarithm of the quantity of funds. Banks' funding cost is increasing in q and is decreasing in z , that is, $r_q > 0$ and $r_z < 0$. As a result, a more productive bank can raise more funding for funding cost r . A simple example of the funding cost function is $r(q|z) = \frac{1}{\theta}(q - z)$, where $\theta \equiv \frac{1}{r_q}$ is the semielasticity of funding supply. Banks then lend to firms with a rate R . For now, the distribution of the productivity z and the lending rate R are taken as exogenous. We will endogenize these variables in Section 2.2.

Banks face a size-based regulation that classifies banks into $I + 1$ categories based on I size thresholds, q_i , where $i = 1, \dots, I$. If a bank's assets cross threshold q_i , it will incur an additional regulatory cost that is equivalent to a τ_i fraction of its profits.¹⁹ Our approach of measuring regulatory costs as a tax is in a spirit similar to Posner (1971). One can translate the regulatory costs to an annual monetary value by multiplying them by the annual profits. One can also interpret τ_i as the fraction of bank value loss due to the present value of all future regulatory costs. This formulation of regulatory costs can be applied in many other settings in which the specifics of regulations may differ.

Facing the size-based regulation, banks choose size q to maximize profits

$$\max_q \pi(q|z) = \max_q (R - r(q|z)) \exp(q) \cdot \prod_{i=1}^I (1 - \tau_i \mathbb{1}_{q \geq q_i}). \quad (1)$$

As shown in the above formula, the regulatory costs increase discretely by τ_i once a bank crosses the i 's threshold. τ_i captures the regulations that come into effect at the corresponding threshold. Note that τ_i does not include regulations that all banks need to comply with or regulations triggered by other thresholds. Nor does τ_i capture the potential indirect costs of regulation on the rest of the economy, which we will discuss in Section 2.3.

Certain regulations may generate private value for banks in compliance with them. For instance, regulatory disclosure can reduce the cost of capital for banks above the threshold. In this case, the tax rate reflects a net regulatory cost, which is the difference between the reporting burden and the private value from a lower cost of capital. We do not attempt to separate further the reporting

¹⁹ In the data, bank profits may fluctuate because of random shocks, but the regulatory burden is largely independent of these random fluctuations. Therefore, one should interpret the regulatory cost as a τ_i fraction of the expected profits rather than realized profits.

burden and the private value to the regulated parties because it is sufficient to estimate the net costs for CBA. Regulators can compare the estimated net costs borne by financial institutions with the social benefits of regulation.

We first solve for the optimal undistorted log assets when there is no regulation by setting all τ_i to zero:

$$q_0(z) \equiv \arg \max_q (R - r(q|z)) \exp(q). \quad (2)$$

The optimal undistorted size is obtained by equalizing the profit margin to the inverse semielasticity of funding supply, $R - r(q_0(z)|z) = r_q(q_0(z)|z)$. If the funding supply function is $r(q|z) = \frac{1}{\theta}(q - z)$, then the optimal undistorted size is given by

$$q_0(z) = z + \theta R - 1, \quad (3)$$

where θ is the constant semielasticity of funding supply.

Now we solve for banks' optimal size when the regulation is present.²⁰ Regulation creates a discrete jump in regulatory costs as represented by the red dotted line in Figure 3, panel A. Banks can avoid the regulation by keeping their assets below the threshold. However, this strategy is costly because banks give up profits they could have earned if they operated on their undistorted scale. The cost of staying below the threshold increases with the productivity z , as more-productive banks need to give up more assets to do so, as represented by the blue dashed line in Figure 3, panel A. As shown in Figure 3, panel B, banks just above the regulatory threshold find it more profitable to stay below the threshold because they only need to shrink by a little bit. Banks whose undistorted size far exceeds the regulatory threshold find it more profitable to operate at their undistorted scale. Formally, we can derive the optimal asset choice as a function of productivity:

$$q^*(z) = \begin{cases} \underline{q}_i & z \in [\underline{z}_i, \bar{z}_i] \\ q_0(z) & z \notin [\underline{z}_i, \bar{z}_i] \end{cases}, \quad (4)$$

where $\underline{z}_i$ is the productivity of a bank whose undistorted assets are equal to the regulatory threshold

$$\underline{q}_i = q_0(\underline{z}_i), \quad (5)$$

and $\bar{z}_i$ is the productivity of a marginal bank that is indifferent between staying below the threshold or paying the regulatory costs,

$$\bar{q}_i \equiv q_0(\bar{z}_i). \quad (6)$$

The indifference condition of the marginal bank is given by

$$(R - r(\bar{q}_i|\bar{z}_i)) \exp(\bar{q}_i)(1 - \tau_i) = (R - r(\underline{q}_i|\bar{z}_i)) \exp(\underline{q}_i). \quad (7)$$

²⁰ We assume the regulatory thresholds are distant enough from each other so that banks only consider whether to avoid the nearest threshold.

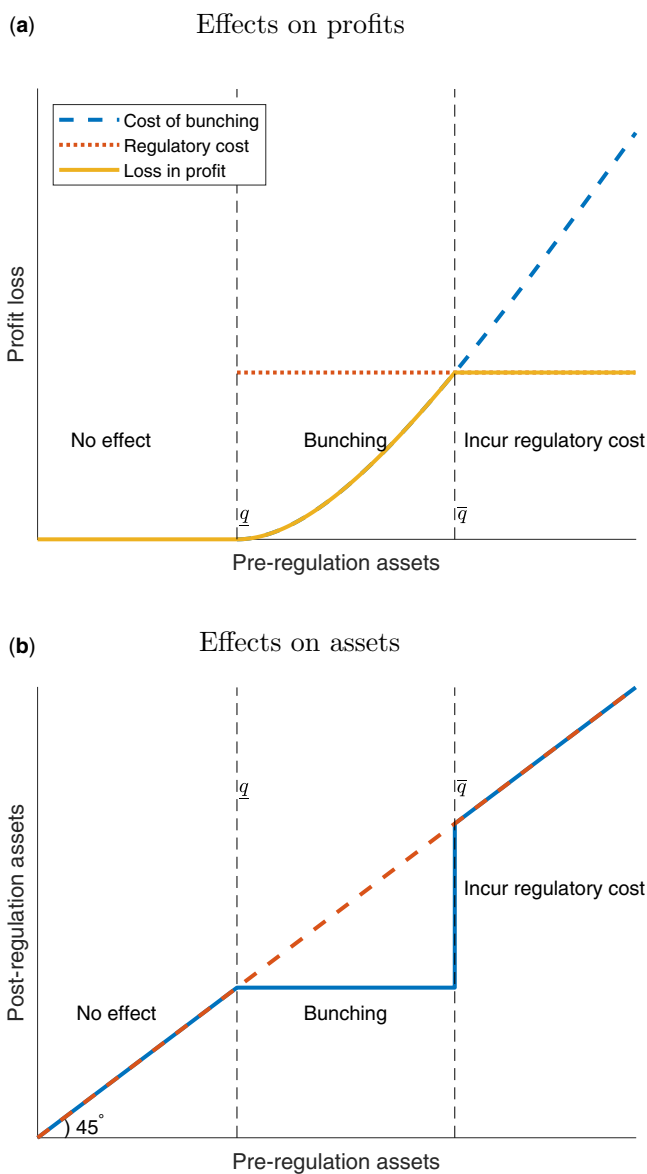


Figure 3

Effects of threshold-based regulation on bank assets and profits

The top panel shows regulatory costs and costs of staying below the threshold as functions of preregulation assets. The bottom panel shows the postregulation assets as a function of preregulation assets.

Banks with productivity between $\underline{z}_i$ and $\bar{z}_i$ bunch at the regulatory threshold, $\underline{q}_i$, while banks outside the bunching interval operate in their optimal scale.²¹ Using the above indifference condition of the marginal bank, Equation (7), we can derive a sufficient statistic formula (Chetty 2009) for the regulatory costs.

Proposition 1. The direct regulatory cost τ_i that comes into effect at threshold i is given by the following sufficient statistic formula:

$$\tau_i \simeq 1 - \left(\bar{q}_i - \underline{q}_i + 1 \right) \exp \left(\underline{q}_i - \bar{q}_i \right), \quad (8)$$

where the approximation is exact when the funding supply function has a constant semielasticity.

Proof. See Appendix A.2. ■

Equation (8) shows that the regulatory cost τ_i only depends on the difference between the marginal bank's log assets, $\bar{q}_i$, and the regulatory threshold, $\underline{q}_i$. It does not depend on the lending and deposit rates, R and r . The intuition is the following. Banks trade off the costs of regulation and the costs of staying below the threshold, which are expressed as a percentage of banks' profits. Although a higher lending rate (or a lower deposit rate) increases banks' profits, it does not change the relative magnitude of regulatory costs and the costs of staying below the threshold. In other words, if our goal is to estimate the tax-equivalent costs of regulation borne by banks, we only need the size distribution of banks as the data input. However, if we would like to translate the regulatory costs to a dollar value, we would need the lending and deposit rates to compute banks' profits.

It is worth noting that sufficient statistic formula (8) still works when banks have heterogeneous semielasticities.²² Intuitively, because the regulatory cost is modeled as a proportional tax, the semielasticity parameter is canceled out from the indifference condition, Equation (7). Heterogeneous semielasticities matter only when we calculate the dollar value of regulatory costs. Such heterogeneity can be captured by multiplying the proportional tax rate with bank-specific profits.

2.2 Firms

We now endogenize the lending rate by introducing the firm sector, which borrows capital from banks and produces output using a Cobb-Douglas

²¹ In our current model, banks choose their size to avoid regulation. It is equivalent to recasting the model into one where banks choose their funding rates to avoid regulation. Specifically, because the funding supply is upward sloping, banks can pay a low rate so that their size becomes smaller.

²² See Appendix A.3 for detailed derivation.

production function:

$$\max_K \Pi = AK^\alpha - RK, \quad (9)$$

where $Y = AK^\alpha$ is the output, K is the capital, and A is the total factor productivity.

The aggregate supply of capital is given by summing the credit supply among banks in the economy:

$$K^s(R) \equiv N \int \exp(q^*(z|R))g(z)dz, \quad (10)$$

where N is the number of banks, $g(z)$ is the probability density function of banks' productivity, and $q^*(z|R)$ is banks' optimal size choice defined in Equation (4).

The equilibrium lending rate R is determined by the market-clearing condition in the lending market:

$$K^s(R) = \left(\frac{R}{A\alpha} \right)^{\frac{1}{\alpha-1}}. \quad (11)$$

2.3 Bank entry and the distribution of productivity

We endogenize the equilibrium distribution of banks' productivity $g(z)$ by introducing growth, entry, and exit. Banks' productivity z evolves following a Brownian motion: $dz_t = \mu_z dt + \sigma_z dB_t$. The value function v of a bank with a current productivity z_0 is defined by

$$v(z_0) \equiv \mathbb{E} \left[\int_0^\infty e^{-(\rho+\lambda)t} \pi(q^*(z_t)|z_t) dt \mid z_0 \right], \quad (12)$$

where ρ is the discount rate, λ is the exogenous exit rate, and π represents the profits resulting from banks' optimal size choice as defined by Equation (1).

New banks endogenously enter the economy. The entry rate is given by the following condition:

$$m = \bar{m} \exp \left(\eta \left(\int v(z) \psi(z) dz - c_e \right) \right), \quad (13)$$

where c_e is the entry costs, $\psi(z)$ is the distribution of potential entrants' productivity, η is the entry elasticity, and $\bar{m}$ is the long-run entry mass when the expected value of entry equals entry costs. For simplicity, we assume the new entrants have the same starting productivity of z_n . The distribution of the productivity evolves according to the following Kolmogorov forward equation:

$$\frac{\partial g(z,t)}{\partial t} = -\frac{\partial}{\partial z} [\mu_z g(z,t)] + \frac{1}{2} \frac{\partial^2}{\partial z^2} [\sigma_z^2 g(z,t)] - \lambda g(z,t) + \frac{m}{N} \psi(z). \quad (14)$$

An equilibrium exists in this economy, which is defined by bank value function $v(z)$, the probability density function of productivity $g(z)$, the number

of banks N , the equilibrium lending rate R , and the aggregate capital K such that:

1. Incumbent banks optimally choose their credit supply given by Equation (4).
2. Potential entrants choose to enter the economy according to Equation (13).
3. Firms optimally choose their credit demand given by Equation (9).
4. The lending market clears.
5. The distribution of banks reaches equilibrium $\frac{\partial g(z,t)}{\partial t} = 0, \forall z$.

Assuming the semielasticity of the funding supply is a constant, we can show that bank assets in the equilibrium follow a power-law distribution.²³ The intuition is that bank assets follow proportional random growth in the absence of regulatory distortion, which generates a power-law distribution in the equilibrium (Gabaix 2016).

2.4 Indirect cost of regulation

We define the indirect costs of regulation, $\tau_{indirect}$, as the percentage output change between the regulated and unregulated economy:

$$\tau_{indirect} = \frac{Y(0) - Y(\tau)}{Y(0)}. \quad (15)$$

where $Y(\tau)$ and $Y(0)$ are the equilibrium firm output in the regulated and unregulated economies, respectively:

$$\begin{aligned} Y(\tau) &= A \left(N(\tau) \int \exp(q^*(z|R(\tau), \tau)) g(z|\tau) dz \right)^\alpha, \\ Y(0) &= A \left(N(0) \int \exp(q^*(z|R(0), 0)) g(z|0) dz \right)^\alpha. \end{aligned} \quad (16)$$

The above equation shows that the indirect costs of regulation can arise from both the intensive and extensive margins. First, regulation reduces the credit supply from incumbent banks q^* (intensive margin). Second, regulation can affect banks' entry incentive, and consequently, the total number of banks N (extensive margin). Note that the sign of the second effect is ambiguous. On the one hand, as regulation reduces the incumbent banks' credit supply, the incentive to enter may increase. On the other hand, because new entrants may eventually grow bigger and face the regulation, heightened regulation may reduce their incentive to enter. Therefore, whether regulation increases or decreases bank entry is an empirical question.

²³ See Appendix A.4 for the derivation of the equilibrium distribution.

3. Estimation

We estimate the model in two stages. First, we estimate the direct regulatory costs using a maximum likelihood estimator. Second, we estimate the indirect regulatory costs by comparing the total output of bank-dependent firms in counterfactual economies with and without regulation. Separating the estimation into two stages clarifies what assumptions and data are needed to quantify the direct and indirect costs of regulation. Estimating the direct costs of regulation only requires the data on the size distribution of banks, as shown in Section 2.1. However, estimating the indirect costs requires stronger assumptions and more parameters on the production functions of banks and firms. We calibrate these additional parameters to either the values in the prior literature or the corresponding moments in the data. We also discuss how these assumptions can affect the estimates of indirect costs in Section 3.4.

Note that we elect not to use the changes in bank values or bank entries before and after the Dodd-Frank Act to estimate regulatory costs. Although these two moments are related to the regulatory burden on banks, they can be driven by an ultra-low-interest-rate environment, market perception of risks, or the size of the too-big-to-fail subsidy, among other factors. Instead, we use the distortion in the size distribution of banks around the Dodd-Frank thresholds, which is unlikely to be driven by other factors as discussed in Section 1.2. Using the estimated regulatory costs from the bunching patterns, we can then quantify how much of the reduction in market valuation of bank shares and the entry rate can be explained by the regulation.

3.1 Data

We combine two main data sources: the Consolidated Reports of Condition and Income (Call Reports) for commercial banks that are not part of a bank holding company and the Consolidated Financial Statements for Holding Companies (FRY-9C reports) for bank holding companies. Since the Dodd-Frank Act applies to the highest holding entity, we only consider the total consolidated assets for the estimation.²⁴ We thus refer to a standalone commercial bank and a bank holding company as a bank. The sample period is 2001–2019. We exclude from the sample banks with assets less than \$1 billion.

Table 1 provides the summary statistics for our sample. The sample covers around 40,000 bank-quarter observations. The average asset size is \$28 billion. The mean and standard deviation of the annual asset growth rate are 7.7% and 8.7%, respectively. The average profits per dollar of assets is 1.6%. We also report a set of regulation-related administrative expense items, including legal, data processing, advisory, printing and supplies, auditing, communications, and labor expenses. Note that we do not use these expense items in our structural estimation because these expense items do not capture all regulatory

²⁴ See article 12 CFR Â§252.13.

Table 1
Summary statistics

	N	mean	sd	p5	p25	p50	p75	p95
Assets	39228	27.905	163.363	1.056	1.376	2.232	5.845	81.776
Assets growth rate	33616	7.655	11.504	-7.081	1.484	5.809	11.632	29.310
Assets growth rate (sd)	38578	8.668	5.071	2.385	4.736	7.663	11.719	18.100
Profits	39228	1.624	1.328	0.223	1.074	1.503	1.948	3.063
Total admin expenses	39228	0.201	0.184	0.000	0.083	0.173	0.278	0.517
Legal	39228	0.026	0.051	0.000	0.000	0.000	0.037	0.129
Data processing	39228	0.101	0.110	0.000	0.000	0.083	0.149	0.295
Advisory	39228	0.028	0.055	0.000	0.000	0.000	0.039	0.132
Printing and supplies	39228	0.015	0.027	0.000	0.000	0.000	0.029	0.065
Auditing	39228	0.012	0.024	0.000	0.000	0.000	0.022	0.059
Communications	39228	0.013	0.025	0.000	0.000	0.000	0.025	0.059
# employees	39228	0.217	0.106	0.070	0.148	0.206	0.271	0.398
Salaries	39228	1.618	0.812	0.736	1.235	1.514	1.804	2.705

This table reports the summary statistics of all the U.S. banks from 2001 to 2019. We use the highest-level ownership as the observation unit because the Dodd-Frank Act applies to both the bank level and bank holding company (BHC) level. The data are quarterly. Assets are reported in billions. Profits and expenses are annualized and reported as a percentage of the assets. The number of employees is reported per millions of assets.

costs. Instead, we will use these expense items when we compare our structural method with other methods of estimating regulatory costs in Section 3.5. In our sample, the average administrative expenses are 0.2 cents per dollar of assets. The number of employees per million of assets is 0.2. The average salaries are 1.6 cents per dollar of assets.

3.2 Stage 1: Direct regulatory costs

We now illustrate how to estimate regulatory costs from the distortion in the size distribution. Following Garicano, Lelarge, and Van Reenen (2016), we assume log assets are observed with a structural error, $a = q + u$, which follows a normal distribution: $u \sim N(0, \sigma^2)$. This structural error accounts for any empirical departures from the model. Specifically, the theoretical model implies a sharp bunching exactly at regulatory thresholds. However, the bunching pattern is more diffused in the data because (1) the empirical measure of bank size may not correspond to the exact definition of bank size for all the regulatory provisions, and (2) banks face random deposit flows and fluctuations in asset values.

We further assume the undistorted assets follow a power-law distribution: $\exp(q) \sim c \exp(q)^{-\beta}$. This assumption is consistent with the equilibrium property of our model and can be verified in the data. Specifically, Figure 4, panel A, plots the log frequency versus the log size of banks in the pre-Dodd-Frank sample. The relationship is close to a straight line, which suggests that the distribution can be well approximated by a power-law (Gabaix 2016).²⁵ We will examine the robustness of the results to this assumption in Section 4. We also assume that the semielasticity of funding supply is constant locally

²⁵ Internet Appendix Table OA.1 conducts goodness-of-fit tests for the distribution assumption and finds the power-law distribution fits well to the data.

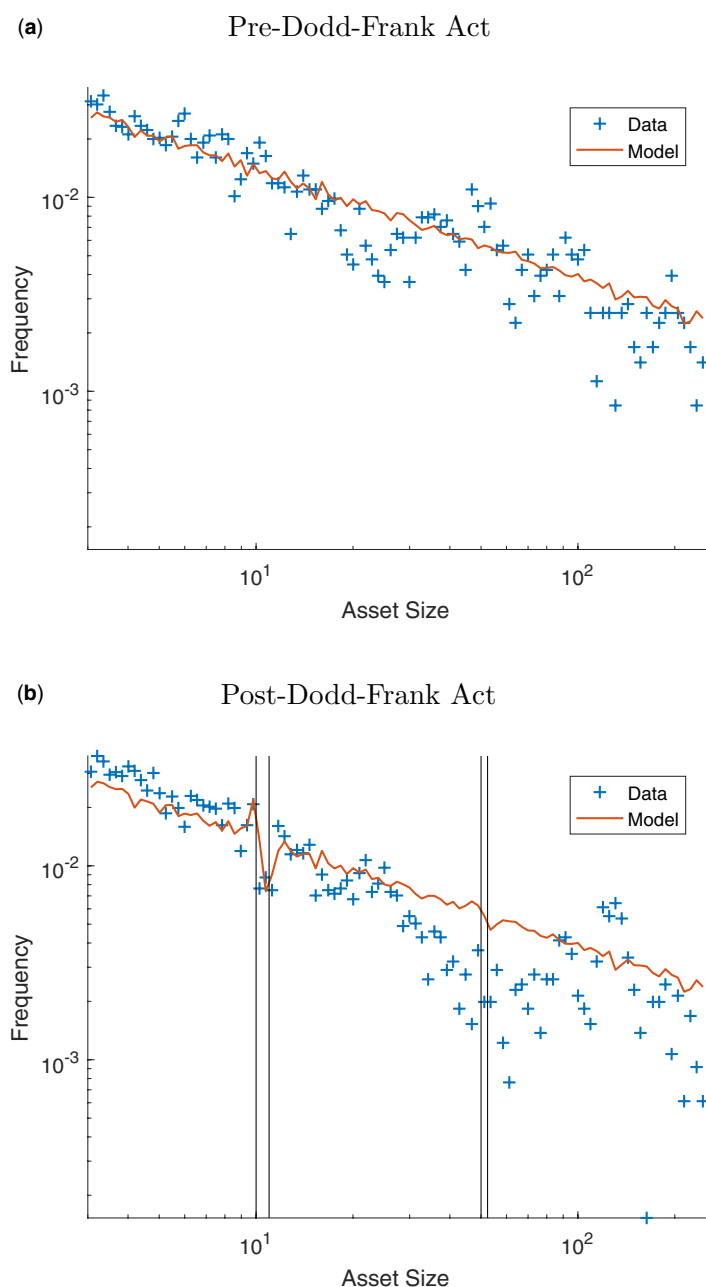


Figure 4
The size distribution of banks: Data versus model

This figure shows the probability density of bank size in the model and in the data before and after the Dodd-Frank Act, respectively. Both axes are in log scale. The vertical lines represent the bunching range around the \$10 billion and \$50 billion thresholds. Bank size is measured by quarter-end assets. The sample period is from 2010Q3 to 2018Q2 for the post-Dodd-Frank period, and from 2001Q1 to 2007Q3 for the pre-Dodd-Frank period.

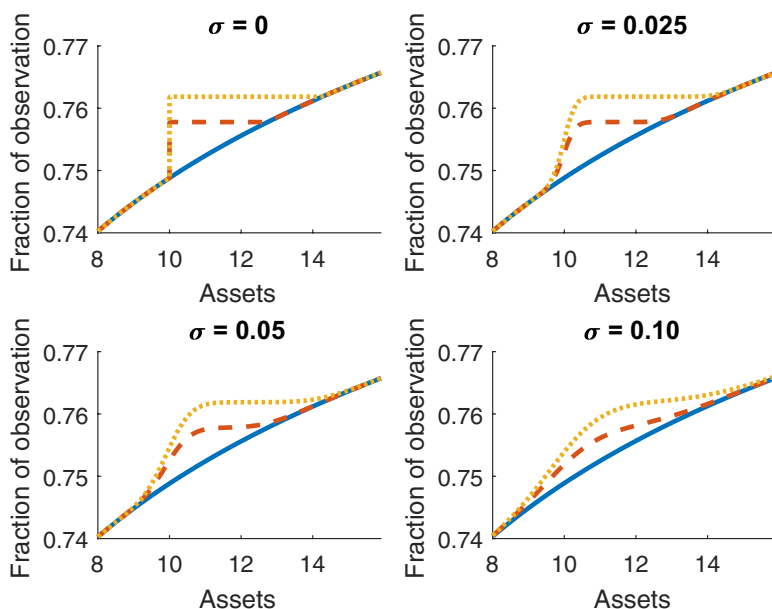


Figure 5
Theoretical size distribution around regulatory thresholds

This figure shows theoretical cumulative distributions of bank size under different values of regulatory costs τ and standard deviation of the structural errors σ . The solid, dashed, and dotted lines represent the case of $\tau = 0$, $\tau = 0.025$, and $\tau = 0.05$, respectively.

around the threshold so that Equation (8) holds exactly. We will examine the magnitude of the approximation error in Section 4.4.

We estimate the regulatory cost parameter using a maximum likelihood estimator:

$$\max_{\Theta} \mathcal{L}(\Theta) = \sum_{j=1}^J \ln f(a_j | \Theta), \quad (17)$$

where f is the likelihood function derived in Appendix A.5, a is the observed log assets, $\Theta = (\tau, \beta, \sigma)$ is the set of unknown parameters, τ is the direct regulatory cost, and σ is the standard deviation of the structural error. β determines the curvature of the distribution of undistorted assets. Note that the scale parameter c of the power-law distribution enters the likelihood as a constant. Therefore, it is unidentified by the data.

Figure 5 illustrates the intuition of the estimation. We simulate the size distribution of banks around the regulatory threshold for different regulatory costs and standard deviations of the structural error. A larger regulatory cost τ leads to a larger abnormal mass around the regulatory threshold. A larger standard deviation of the structural error σ makes the abnormal mass more diffused around the threshold. The power-law exponent, β , is pinned down by

Table 2
Maximum likelihood estimation of regulatory costs

<i>A. \$10 billion threshold</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.001]
σ	Measurement error volatility (in %)	4.258	[0.386]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	10.973	[0.086]
τ	Cost of regulation (% of profit)	0.405	[0.066]
<i>B. \$50 billion threshold</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.083	[0.002]
σ	Measurement error volatility (in %)	2.290	[0.498]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	52.393	[0.517]
τ	Cost of regulation (% of profit)	0.106	[0.046]

This table presents the results of the maximum likelihood estimation of the regulatory costs using the power-law distribution. The sample period is from 2010Q3 to 2018Q2. The estimation sample of panel A includes banks with assets between \$3 billion and \$40 billion. The estimation sample of panel B includes banks with assets above \$40 billion. Standard errors are obtained via the inverse of the Hessian matrix.

the global shape of the distribution, that is, observations far away from the regulatory threshold. The goal of the maximum likelihood estimation is to find a set of parameters that can generate a similar distortion to what is observed in the data.

We conduct the estimation for each regulatory threshold separately using subsamples of banks around the thresholds. Doing so is sensible because the thresholds are far apart, and the bunching ranges are unlikely to overlap. We use banks in the \$3 billion to \$40 billion interval for the \$10 billion threshold, and banks above \$40 billion for the \$50 billion threshold.²⁶ The sample period is from 2010Q3 to 2018Q2.²⁷

We present the result of the maximum likelihood estimation in Table 2. The point estimate for the annual regulatory costs triggered by the \$10 billion threshold is 0.41% of average annual profits. The estimate is statistically significant with a standard error of 0.066%. Note that this estimate captures the additional regulatory cost imposed by the Dodd-Frank Act that is triggered at the \$10 billion threshold. It does not include the regulations already in place before the Dodd-Frank Act. Given this estimate, we calculate the undistorted assets of the marginal bank, $\exp(\bar{q})$, to be around \$11 billion using Equation (8). Next, in the second panel of Table 2 we examine the \$50 billion threshold. The point estimate of additional regulatory costs when crossing the \$50 billion threshold is 0.11%. Adding the 0.41% regulatory costs triggered by crossing the \$10 billion threshold, the total regulatory cost imposed by the Dodd-Frank Act on banks above \$50 billion is 0.52%. We also calculate that the undistorted assets of the marginal bank at this threshold are around \$52 billion.

²⁶ The results are robust to alternative sample ranges, as shown in the Internet Appendix Table OA.2.

²⁷ The estimation is robust to adding pre-Dodd-Frank data, as shown in Internet Appendix Table OA.3.

Table 3
Calibrated parameters for policy experiment

Parameter	Value	Target	Data	Model
μ_z : productivity growth	7.700	Asset growth	7.700	7.700
σ_z : productivity diffusion	8.700	Asset growth std	8.700	8.700
λ : exit rate	4.400	Exit rate	4.400	4.400
θ : semielasticity of funding supply	61.728	Profit margin	1.620	1.620
ρ : discount rate	7.000	Market-to-book	1.100	1.228
z_n : productivity of new entrants	-3.470	Size of entrants	0.250	0.239
A : total factor productivity	8.000	Lending rate	5.000	4.924
c_e : entry costs	0.120	Average bank size	19.067	21.781
$\bar{m}$: mass of entry in the long run	1.000	Normalize to one	-	-
α : curvature of production function	0.300	Cooper et al. (2007)	-	-
η : elasticity of entry	100.000	Moll (2018)	-	-
τ_{10} : regulatory cost at \$10B	0.410	Table 2	-	-
τ_{50} : regulatory cost at \$50B	0.110	Table 2	-	-

Columns 1 and 2 of this table report the calibrated parameters for the counterfactual simulations. Columns 3–5 report the targeted moments in the data and their model counterpart.

To put this number into perspective, we can calculate the dollar costs of the regulatory burden imposed by the Dodd-Frank Act on a bank with \$50 billion in assets. Given the profits of around 1.6% of total assets (Table 1), our estimate implies a dollar value of regulatory costs around \$4.16 million per year. Our estimate is equivalent to the annual expense of hiring 52 additional compliance officers, assuming the average annual compensation for a compliance officer is around \$80,000 (Feldman, Heinecke, and Schmidt 2013). This estimate is also related to Kisin and Manela (2016), who estimate the shadow cost of bank capital requirements before the 2008 financial crisis to be around 0.4% of banks' annual profits. Although the exact regulations are different, the estimates are similar in order of magnitude.

3.3 Stage 2: Indirect regulatory costs

So far, we have estimated the direct costs of regulation on banks. We now calibrate other model parameters that affect the indirect costs of regulation on firms that borrow from banks. We calibrate these parameters using the corresponding moments in the data or the values in the prior literature.

Table 3 presents the parameters used for counterfactual policy experiments. We first discuss the directly specified parameters. We set the regulatory costs triggered at the \$10 billion and \$50 billion thresholds to 0.41% and 0.11% according to our estimates in Table 2. We normalize the mass of entry in the long-run $\bar{m}$ to 1. The curvature of the production function α is set to 0.3, as in Cooper, Haltiwanger, and Willis (2007). This parameter determines the aggregate elasticity of credit demand to the lending rate. We set the elasticity of entry η to a large number, 100, so that the equilibrium approximates the competitive entry in the long run (Moll 2018).

Next we discuss parameters calibrated to our data, for which we also report the targeted data moments and their model counterparts. The drift of productivity, μ_z , is calibrated to match the average asset growth rates in the

Table 4
Model fit: Untargeted moments

Moments	Pre Dodd–Frank		Post Dodd–Frank	
	Data	Model	Data	Model
Share of banks: small banks	89.609	87.787	89.318	87.777
Share of banks: medium banks	6.569	7.223	7.065	7.230
Share of banks: big banks	3.822	4.990	3.617	4.993
Share of assets: small banks	10.385	5.714	9.557	5.711
Share of assets: medium banks	10.472	7.327	7.556	7.311
Share of assets: big banks	79.142	86.958	82.887	86.978
Imbalance ratio: \$10 billion	1.091	0.884	0.421	0.564
Imbalance ratio: \$20 billion	1.182	0.902	0.895	0.896
Imbalance ratio: \$30 billion	1.444	1.032	1.240	0.991
Imbalance ratio: \$40 billion	0.833	0.886	1.045	0.864
Imbalance ratio: \$50 billion	0.710	0.993	0.458	0.661

This table reports the untargeted moments in the data and their model counterpart. Banks are categorized as small (<\$10 billion), medium (\$10 billion to \$50 billion), and big (>\$50 billion). The imbalance ratio is defined as the number of banks in a narrow interval above the threshold divided by the number of banks in a narrow interval below the threshold (e.g., \$10 billion–\$11 billion vs. \$9 billion–\$10 billion). We use a \$1 billion window for the \$10, \$20, and \$30 billion thresholds, and a \$2 billion window for the \$40 and \$50 billion thresholds. The choice of the window size is consistent with the estimated marginal bank in Table 2.

pre-Dodd-Frank sample, which is 7.7%. The diffusion of productivity, σ_z , is calibrated to match the standard deviation of asset growth, which is 8.7%. We match the average and the standard deviation of asset growth rates exactly because of a one-to-one correspondence between assets and productivity in the absence of regulation in the model. The exit rate λ is calibrated to match the value observed in the data of 4.4%.²⁸ We assume the funding supply function is $r(q|z) = \frac{1}{\theta}(q - z)$ and calibrate the semielasticity of funding supply θ to match the average profit margin, which is 1.62% in the data. The subjective discount rate of banks is calibrated to match the average ratio of the market value of assets to the book value in the pre-Dodd-Frank data. The productivity of new entrants, z_n , is calibrated to match the size of new entrants observed in the data, which is \$0.25 billion. Total factor productivity, A , is calibrated to match an average lending rate of 5%. Intuitively, higher total factor productivity drives up the demand for credit and the equilibrium lending rate. The entry cost, c_e , is calibrated to match the average bank size in 2010Q3. Overall, the model does a reasonable job of matching most of the targeted moments, as shown in the second column of Table 3.

Next, we examine the quantitative fit of the model for untargeted moments in Table 4. We group banks in three size categories: small (<\$10 billion), median (\$10 billion to \$50 billion), and big (>\$50 billion). We calculate the share of banks and assets in each size category in both the data and the model for the pre-Dodd-Frank period (columns 1 and 2) and for the post-Dodd-Frank period (columns 3 and 4). We find that the model generates a highly skewed size distribution. For instance, big banks account for 5% of banks but 87% of

²⁸ This value includes exits due to bank failures and due to mergers and acquisitions. In Internet Appendix Table OA.4, we evaluate the robustness of our results to alternative calibrations of the exit rates.

assets in the model. The corresponding moments in the data are 4% and 80%, respectively.

We further examine whether the model can match the bunching pattern around the regulatory thresholds. We construct an imbalance ratio, defined as the number of banks in a narrow interval above the threshold divided by the number of banks in a narrow interval below the threshold (e.g., \$10 billion–\$11 billion vs. \$9 billion–\$10 billion).²⁹ A reduction in the imbalance ratio indicates that the number of banks above the threshold becomes scarcer relative to those below the threshold. In the data, the imbalance ratio at \$10 billion has decreased from 1.09 to 0.42 from the pre- to the post-Dodd-Frank samples. The model generates a similar decrease in the imbalance ratio. A similar pattern can be found for the \$50 billion threshold. Finally, the model also matches the imbalance ratios at other round numbers that are not regulatory thresholds. As expected, imbalance ratios do not systematically decrease for nonregulatory thresholds.

Figure 4 compares the model-predicted probability density function with that in the data. The model fits the data quite well over the entire distribution. We further zoom in to the region around the regulatory thresholds in Figure 6. Note that the probability density function is quite noisy in small samples. Consequently, we plot the cumulative distribution function instead. We find that the model generates similar departures from the power-law distribution to the data.

We use the calibrated model to quantify the indirect cost of regulation. We first simulate a baseline economy without the Dodd-Frank Act in column 1 of Table 5. Then we simulate an economy with the Dodd-Frank Act and present the percentage change with respect to the baseline economy in column 2 of Table 5. We first examine how the Dodd-Frank Act affects the total number of banks in the equilibrium. In theory, the Dodd-Frank Act can have two countervailing forces on bank entry. On the one hand, the regulation increases the profits of potential entrants because incumbent banks around the regulatory thresholds reduce their lending to avoid the regulation. On the other hand, the prospect of facing tightened regulation in the future for the potential entrants reduces their incentive to enter. The simulation suggests that the latter force dominates. The total mass of banks decreases by 0.18% after the introduction of the Dodd-Frank Act. This prediction is consistent with the falling entry rates after the Dodd-Frank Act observed in the data, as shown in Figure 7, panel A.

Practitioners often argue that high regulatory costs imposed by the Dodd-Frank Act have led to the decline in bank franchise values, as shown in Figure 7, panel B. Although the decline in franchise values is consistent with heightened regulatory costs, it could be driven by many other factors, including an ultra-low-interest-rate environment (Calomiris and Nissim 2014;

²⁹ We use a \$1 billion window for \$10, \$20, and \$30 billion thresholds, and a \$2 billion window for \$40 and \$50 billion thresholds to be consistent with the bunching range estimated in Table 2.

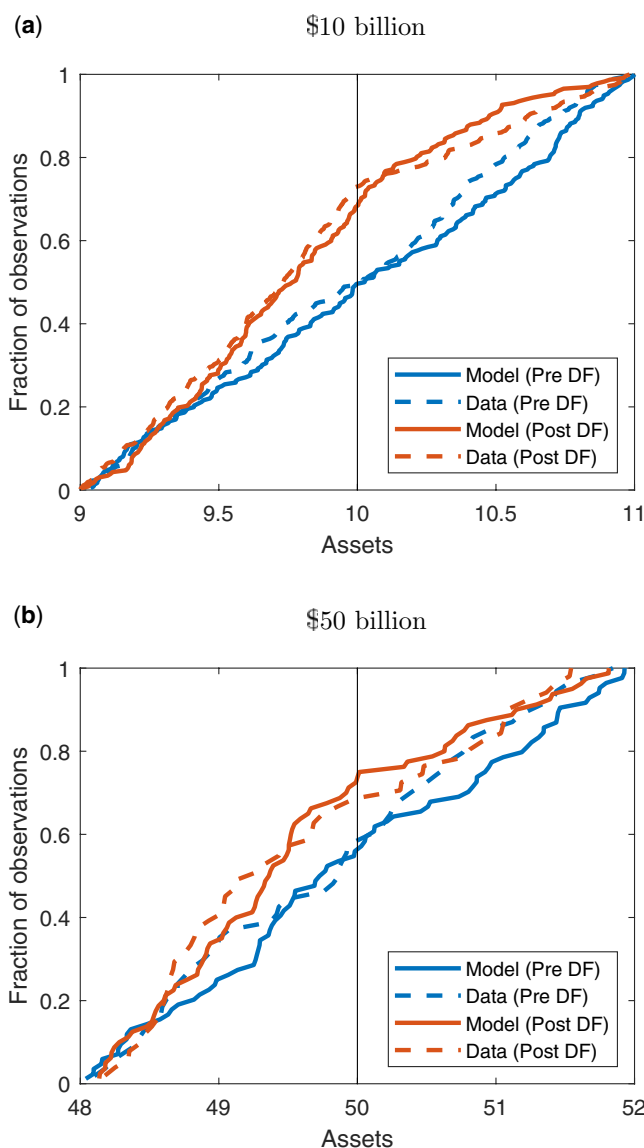


Figure 6
Simulated size distribution around regulatory thresholds

This figure shows the model-simulated and empirical distributions of bank size conditional on the narrow intervals around the \$10 billion and the \$50 billion thresholds before and after the Dodd-Frank Act, respectively.

Whited, Wu, and Xiao 2021), market reassessment of risks (Sarin and Summers 2016), and the removal of too-big-to-fail subsidies (Atkeson, d'Avernas, Eisfeldt, and Weill 2019; Berndt, Duffie, and Zhu 2020). As a result, we abstain

Table 5
Counterfactual simulation

Counterfactual	Baseline	Dodd-Frank	Regulatory relief
Panel (a): All banks			
Mass of banks	1.000	−0.184%	−0.094%
Lending quantity	1.000	−0.065%	−0.032%
Output	0.164	−0.020%	−0.010%
Market-to-book	1.228	−0.221%	−0.104%
Lending rate	0.049	0.046%	0.023%
Panel (b): Annual profits by size group			
Small banks	0.023	0.068%	0.033%
Medium banks	0.358	−0.399%	−0.221%
Big banks	6.150	−1.268%	−0.593%
Panel (c): Asset shares by size group			
Small banks	0.057	−0.061%	−0.034%
Medium banks	0.073	−0.216%	−0.123%
Big banks	0.870	0.022%	0.013%
Panel (d): Shares of banks by size group			
Small banks	0.878	−0.012%	−0.006%
Medium banks	0.072	0.089%	0.042%
Big banks	0.050	0.075%	0.042%

This table reports the results of counterfactual simulations. Column 1 reports the equilibrium quantities and prices in the baseline economy without the Dodd-Frank Act. The mass of incumbent banks and the lending quantity in the baseline economy are normalized to one. The output is expressed per unit of bank lending. Columns 2 and 3 report the simulated economies with the Dodd-Frank Act and the 2018 regulatory relief, respectively. The values are reported as percentage changes with respect to the baseline economy.

from using the change in the bank franchise value to estimate regulatory costs. Instead, we use the estimated regulatory costs to evaluate how much the decline in bank values can be attributed to the Dodd-Frank Act. We find the estimated regulatory costs lead to a 0.22% decrease in bank values, which can only account for 3% of the overall decline in bank franchise value in the postcrisis period. Our estimate suggests nonregulatory factors may have contributed to the decline in bank values.

Next, we investigate the distributional effects of the Dodd-Frank Act in the cross-section of banks. This inquiry is motivated by the fact that big banks are subject to tighter regulation according to the Dodd-Frank Act. We find that big banks indeed suffer a greater decrease in profits than medium and small banks, as shown in panel B of Table 5. Nevertheless, the heightened regulation on big banks does not translate into smaller market shares. Instead, the share of big banks increases in the equilibrium with the Dodd-Frank Act. The reason is twofold. First, medium banks (\$10 billion to \$50 billion) engage in regulatory avoidance, which reduces their average asset size, as shown in panel C. Second, heightened regulatory costs depress bank valuation across the size spectrum, which reduces the entry of small banks (<\$10 billion), as shown in panel D. These results do not imply that the Dodd-Frank Act has worsened the too-big-to-fail problem because we do not explicitly model how regulation affects big banks' risk-taking and their reliance on the implicit government guarantee.

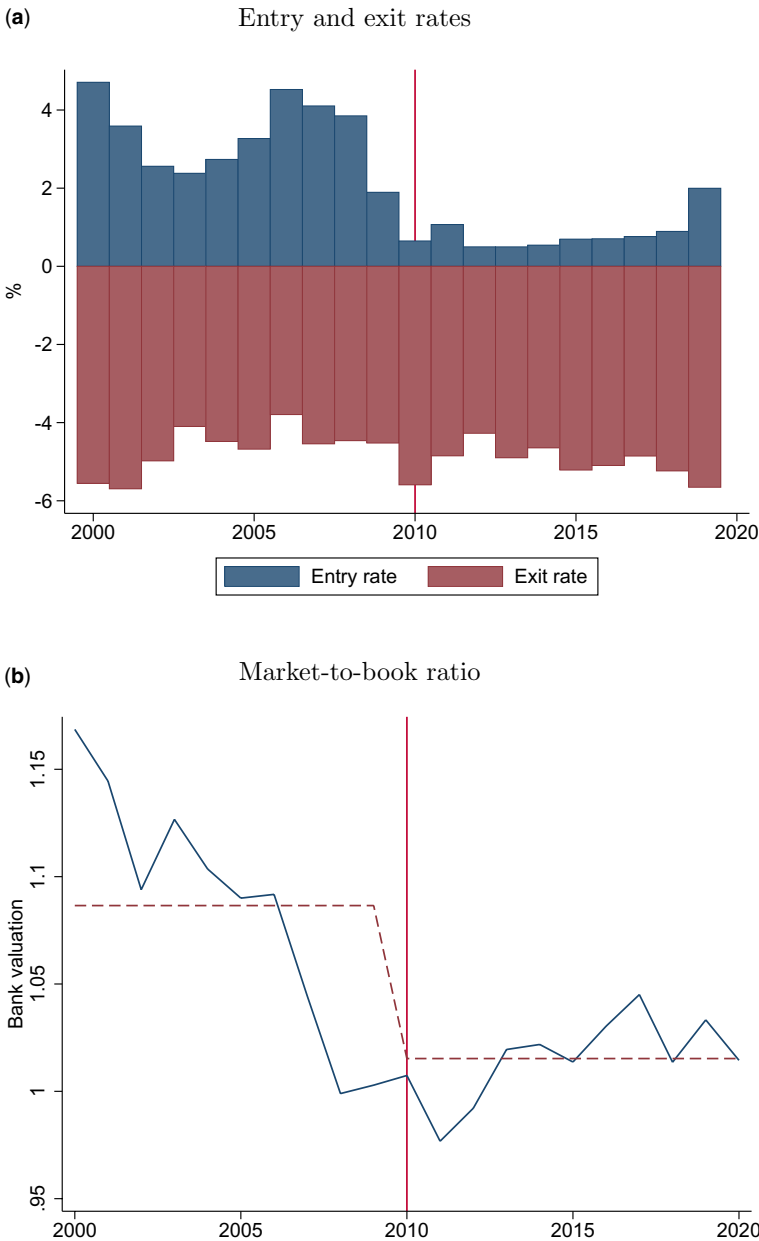


Figure 7
Entry, exit, and bank valuation
Panel A shows entry and exit rates of U.S. banks before and after the Dodd-Frank Act. The entry (exit) rate is calculated by dividing the number of banks that enter the market (drop out from the market) in a year by the total number of banks in that year. Panel B shows the average ratio of market value of assets to the book value of assets for U.S. publicly traded banks.

However, it does suggest a possible unintended consequence of regulation on the bank industry's market structure.

Finally, we examine the indirect costs of regulation on bank-dependent firms. We find that the Dodd-Frank Act increases the equilibrium lending rate by 0.046% and decreases the lending quantity by 0.065%. Overall, the Dodd-Frank Act decreases the total output of bank-dependent firms by around 0.020%. These results suggest that the indirect regulatory costs are modest. One may worry that our estimates of the indirect regulatory costs may be sensitive to the parameters that we use to simulate the counterfactuals. In Table OA.4 of the Internet Appendix, we simulate the baseline and counterfactual economies using alternative values of the discount rate, exit rate, and entry cost. We find the estimated indirect regulatory costs with alternative parameters are in the same order of magnitude as our baseline estimate.

3.4 Discussion of indirect regulatory costs

Although the above exercise provides a novel quantification of the indirect costs of the Dodd-Frank Act, it is also subject to several caveats. First, we assume that banks have the same business model and lend to the same representative firm. Although this assumption simplifies the model, it also prevents us from studying the heterogeneity of indirect costs. In reality, banks and firms exhibit a certain degree of assortative matching (Berger et al. 2005).³⁰ The impact of regulation may be disproportionately felt by some firms, such as those that borrow from the bunching banks.

Second, in the above exercise, we assume that entrants can choose either to be a bank or not to enter the market at all. In reality, entrants can also choose to be a shadow bank. This choice could be important because banks have received heightened competition from shadow banks since the financial crisis and subsequent shifts in the regulatory regime (Buchak et al. 2018; Irani et al. 2021; Begenau and Landvoigt 2021). Introducing shadow banks could affect the estimates in two potential ways. On the one hand, since firms can switch from banks to shadow banks, introducing shadow banks could lower the estimated indirect costs of regulation. On the other hand, the migration of activity to less regulated shadow banks could increase the fragility of the banking system, thus increasing the indirect costs. Quantifying these indirect costs requires a more complex model and data on shadow banks, which we leave for future research.³¹

Finally, the model focuses on the partial equilibrium effects of regulation on the banking sector. The estimated effect on bank lending in this study should

³⁰ See also Strahan (1999) and Chodorow-Reich et al. (2022) for further evidence of bank liquidity provision across firms of different sizes.

³¹ One way to get a sense of how changes in the competitive environment may affect the results in the current theoretical framework is to simulate the effect of the regulatory change in conjunction with an increase in competition by entrants. Table OA.5 shows that the reduction in output is indeed smaller if the regulatory changes are accompanied by a reduction in entry costs.

Table 6
Estimated regulatory costs of Dodd-Frank on banks based on surveys

Source	Sample	Estimate	Date
Bank Director Magazine	Survey of 10 banks	9.9	1/1/2017
American Action Forum	Estimation from Federal Register	1.8	7/1/2016
JPMorgan and Citigroup	Survey of 2 banks	0.9	2012, 2014
Federal Reserve Bank of Minneapolis	Estimation of cost of new hires	1.1	3/1/2013
Bank Director and Grant Thornton LLP Survey	Survey of 130 senior executives	Qualitative	6/1/2013
KPMG 2013 Community Banking Survey	Survey of 100 senior executives	Qualitative	10/1/2013
Florida Chamber Foundation	Survey of 75 banks	Qualitative	7/1/2012
Mercatus Center's Small Bank Survey	Survey of 200 banks	Qualitative	2/1/2014
Risk Management Association Survey	Survey of 230 senior executives	Qualitative	3/1/2013

This table lists surveys on the regulatory costs of the Dodd-Frank Act on banks. For sources with quantitative estimates, we translate the estimates into percentages of net income.

not be interpreted as an estimate of the aggregate impact on the credit supply because firms can substitute loans with other forms of credit, such as issuing corporate bonds (Kashyap, Stein, and Wilcox 1993; Becker and Ivashina 2014). The regulatory impacts may also spill over to households through labor and product markets. Understanding the general equilibrium effects is a much more challenging task, which we leave for future studies.

3.5 Comparisons with existing methods

In this section, we compare our structural estimates with the existing methods of estimating regulatory costs. Note that the cost estimates surveyed in this section are about the direct compliance costs for banks. Therefore, they should be compared with the direct regulatory costs estimated in Section 3.2, which are 0.41%–0.52% of banks' profits.

3.5.1 Survey. One of the most commonly used methods of estimating regulatory costs is conducting surveys. We summarize the surveys on the regulatory costs of the Dodd-Frank Act for banks in Table 6. Many surveys provide only a qualitative indication of high regulatory costs but no quantitative estimates. Among surveys that do provide quantitative estimates, the magnitudes differ greatly. For instance, a survey conducted by *Bank Director Magazine* suggests that the annual regulatory costs are around 9.9% of banks' annual profits. In contrast, a survey conducted by the American Action Forum estimates that regulatory costs are around 1.8%. The survey estimates are usually much larger than the estimates obtained from our revealed preference approach.

3.5.2 Difference-in-differences. Some researchers also use the difference-in-differences approach to estimate regulatory costs. For instance, Hinkes-Jones (2017) estimates regulatory costs by comparing legal and administrative expenses of affected banks with those of unaffected banks. This methodology faces two challenges. The first challenge is measurement. Banks are not required to report legal or other selected noninterest expenses that do not

Table 7
Regulatory costs of the Dodd-Frank Act estimated by difference-in-differences

	(1) # employees	(2) Salaries	(3) Total admin expenses	(4) Legal	(5) Data processing	(6) Advisory	(7) Printing	(8) Auditing	(9) Com- munications
Treat * Post	−0.012 [0.012]	−0.043 [0.081]	−0.012 [0.018]	0.008 [0.008]	0.001 [0.014]	−0.003 [0.008]	0.001 [0.004]	0.000 [0.005]	−0.009** [0.004]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,816	6,816	5,931	3,135	5,067	2,429	2,476	1,323	2,643
Adj. R-squared	0.515	0.039	0.115	0.063	0.006	0.037	0.197	0.043	0.302

This table presents the estimated regulatory costs from the difference-in-difference regression shown in Equation (18). The sample includes U.S. banks with assets between \$3 billion and \$40 billion at the time of the introduction of Dodd-Frank and covers the period of 2003Q1–2018Q2. The dependent variables have been normalized by assets. *Post* is a dummy equal to one for all years after the Dodd-Frank Act (after 2010Q3). *Treat* is a dummy equal to one if the bank’s total consolidated assets are above \$10 billion as of 2010Q2, and zero otherwise. Robust standard errors clustered at the bank level are shown in brackets. All regressions include bank and time fixed effects, and control for the logarithm of the number of bank branches.

exceed defined thresholds that have changed over time. Nevertheless, some banks report them even if they fall short of the thresholds. These issues may create difficulties for researchers who attempt to draw conclusions from trends in these expense items. Furthermore, not all regulatory costs are reflected as expenses in banks’ income statements. For instance, an increased capital requirement is costly but does not directly lead to higher expenses. The second challenge is endogeneity. Because banks can endogenously stay below the threshold and avoid the regulatory costs, a difference-in-differences regression may lead to downward biases.

To illustrate the difficulty of using the difference-in-differences approach to estimate regulatory costs, we apply this methodology to estimate the regulatory costs imposed by the Dodd-Frank Act on banks with more than \$10 billion in assets. The regression model is as follows:

$$Expenses_{i,t} = \alpha_i + \alpha_t + \beta Treat_i \times Post_t + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (18)$$

where *Expenses* includes regulation-related expenses such as legal, data processing, advisory, printing, stationery and supplies, auditing, and communication costs. α_i and α_t are bank and time fixed effects, respectively. $X_{i,t}$ includes the logarithm of the number of branches. The expenses are normalized by assets. We also include the number of employees and total salaries to account for the possibility that regulations may force banks to hire more compliance officers. The sample includes banks with assets between \$3 billion and \$40 billion as of 2010Q2. The sample period is from 2003Q1 to 2018Q2. *Treat* is a dummy equal to one if the bank’s total assets are above \$10 billion as of 2010Q2, and zero otherwise. *Post* is a dummy that equals one for years after the Dodd-Frank Act (after 2010Q3).

Table 7 reports the results. None of the expenses significantly increases for the treated banks. Some expenses, such as the communication expenses, actually have the wrong sign. This result is confirmed by plotting the rolling

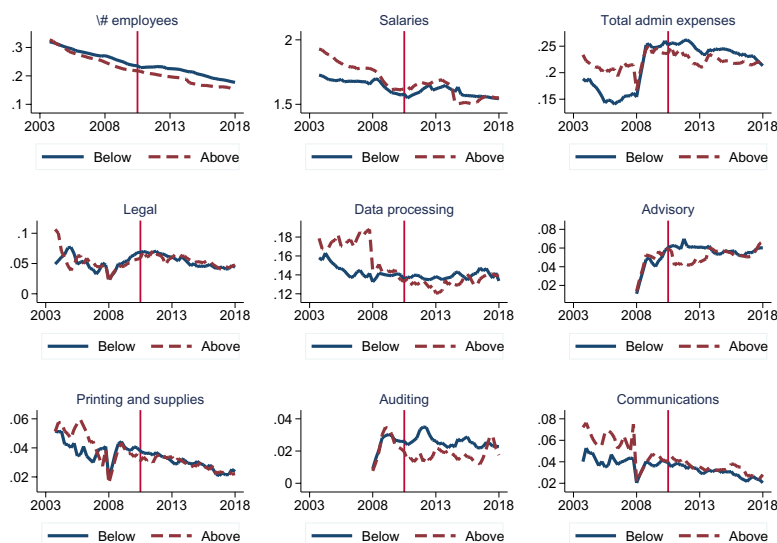


Figure 8

Regulation-related expenses for banks above and below \$10 billion in assets

This figure presents the regulation-related expenses over time. Banks are classified as “below” (“above”) if their total consolidated assets are between \$3 billion and \$10 billion (\$10 billion and \$40 billion) at the start of the Dodd-Frank Act in 2010. The dependent variables are normalized by assets.

four-quarter average annualized expenses for the treated and control groups in Figure 8, which shows no significant increase for the treated group after the Dodd-Frank Act. However, these results should not be interpreted as evidence that the Dodd-Frank Act imposes no costs on banks because of the measurement issue and endogeneity concern discussed earlier.

3.5.3 Regression discontinuity. Another methodology that can be used to estimate regulatory costs is regression discontinuity design. This methodology is subject to the measurement and endogeneity concerns discussed in Section 3.5.2. Nevertheless, we apply this approach to banks around the \$10 billion threshold using the following regression model:

$$Expenses_{i,t} = \beta_0 + \beta_1 \mathbf{1}_{\{Q_{i,t} \geq \underline{Q}\}} + \beta_2 f(Q_{i,t} - \underline{Q}) + \beta_3 \mathbf{1}_{\{Q_{i,t} \geq \underline{Q}\}} f(Q_{i,t} - \underline{Q}) + \varepsilon_{i,t}, \quad (19)$$

where $Expenses_{i,t}$ include the same set of regulation-related expenses as in Section 3.5.2; $Q_{i,t}$ represents the assets; and $f(\cdot)$ is a polynomial function of degree one. β_1 is the increase in the regulation-related expenses when a bank’s assets exceed the regulatory threshold, $\underline{Q}$. The sample period is from 2010Q3 to 2018Q2. Table 8 presents the estimated coefficients, and Figure 9 shows the results graphically. Again, most expenses do not exhibit significant changes above and below the thresholds. There is a small increase in the auditing costs for treated banks and a small decrease in communication costs.

Table 8
Regulatory costs of the Dodd-Frank Act estimated by regression discontinuity

	# employees	Salaries	Total admin expenses	Legal	Data processing	Advisory	Printing	Auditing	Com- munications
Treatment Effect	0.010 [0.024]	0.222 [0.182]	0.025 [0.030]	0.011 [0.021]	0.007 [0.023]	−0.003 [0.015]	−0.005 [0.005]	0.021** [0.010]	−0.010* [0.005]
Observations	559	559	505	256	400	296	203	148	256
Adj. R-squared	0.011	0.004	−0.004	−0.005	0.006	0.008	−0.003	0.028	0.003
Bandwidth	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38

This table presents the estimated regulatory costs from the regression discontinuity design shown in Equation (19). The sample includes U.S. banks with assets between 8.62 billion and 11.38 billion from 2010Q3 to 2018Q2. The dependent variables are normalized by assets. The treatment effect is the increase in the regulation-related expenses when a bank's assets exceed the regulatory threshold. Robust standard errors clustered at the bank level are shown in brackets.

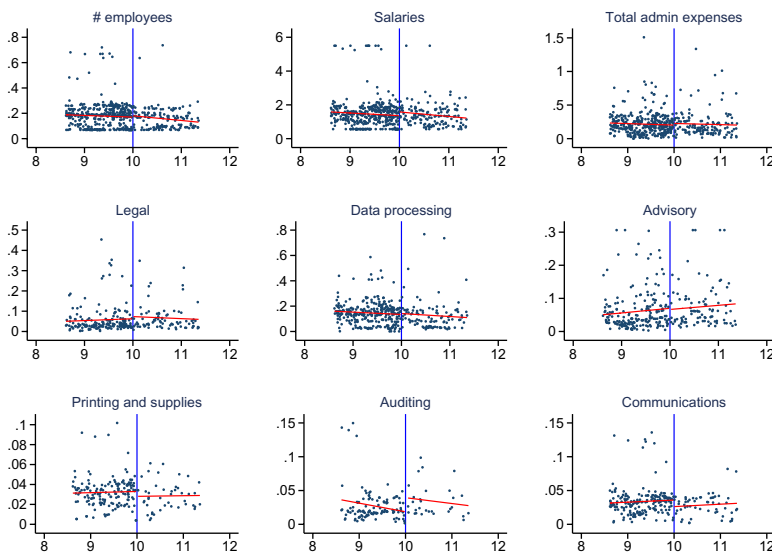


Figure 9
Regulation-related expenses around the regulatory threshold

This figure presents the scatter plot of regulation-related expenses against bank size around the \$10 billion threshold. The sample includes all the bank-quarter observations from 2010Q3 to 2018Q2. The red lines represent the predicted values using the regression of Equation (19). The dependent variables have been normalized by assets.

3.5.4 Summary. The existing methods lead to inconsistent evidence on the regulatory costs of the Dodd-Frank Act. Surveys on banks typically suggest extremely large regulatory costs, consistent with the anecdotal evidence that survey estimates may be inflated to lobby regulators for regulatory relief (Hinkes-Jones 2017; Parker 2018a). In contrast, reduced-form methods such as difference-in-differences and regression discontinuity designs find virtually no evidence of additional regulatory costs due to the Dodd-Frank Act, with the

caveats of the endogeneity and measurement issues. Our structural approach finds a modest level of regulatory costs.

We view our approach as complementary to the existing approaches of evaluating the impacts of regulation. The reduced-form approaches do not require researchers to model the data-generating process. Instead, they require at least some observations to be randomly assigned. However, for big and highly anticipated regulatory changes, regulated parties are likely to engage in regulatory avoidance, so observations are rarely randomly assigned. These selection issues pose challenges for the reduced-form methods. While such selection issues could be addressed by finding valid instruments, it does limit the scope of the analysis to settings in which instruments are available. In comparison, our structural approach allows us to analyze equilibrium responses to big and highly anticipated regulatory changes. With additional assumptions on the data-generating process, we can conduct counterfactual policy experiments and analyze welfare implications.

4. Extension and Robustness

Our revealed preference approach addresses some challenges faced by the existing methods, such as the lack of reliable data and the endogeneity concern. Nevertheless, this approach does require some structural assumptions, and the robustness of the results to these assumptions should be duly assessed. In this section, we conduct a few extensions and robustness checks on our results.

4.1 Regulatory relief in 2018

In 2018, the U.S. Congress passed the Economic Growth, Regulatory Relief, and Consumer Protection Act, partially reversing some regulations of the Dodd-Frank Act. For instance, banks with more than \$10 billion in assets are no longer required to conduct stress tests except for those whose assets are greater than \$250 billion.³² This regulatory relief presents a good validity check for our approach because the estimated regulatory costs should decrease in the postrelief period. Indeed, we find that the estimated regulatory costs at the \$10 billion threshold decrease by around 50% relative to the prerelief value, while the estimated regulatory costs at the \$50 billion threshold almost completely disappear, as shown in Table 9.³³ These estimates are consistent with the visual evidence in Figure 10 that the abnormal densities at the Dodd-Frank regulatory thresholds have significantly reduced after the relief. Note the still-considerable bunching after the regulatory relief because some regulations, such as the Durbin Amendment, remain in place.

³² See “Amendments to the Stress Testing Rule for National Banks and Federal Savings Associations.”

³³ With only one year of data after the regulatory relief, we fix the exponent of the power-law distribution and the structural error volatility to the baseline estimation value as these parameters are quite persistent.

Table 9
Maximum likelihood estimation of regulatory costs: Post-2018 regulatory relief

\$10 billion threshold			
		Estimated value	S.E.
$\exp(\bar{q})$ τ	Assets of marginal bank (\$ Billion)	10.701	[0.159]
	Cost of regulation (% of profit)	0.219	[0.095]
\$50 billion threshold			
		Estimated value	S.E.
$\exp(\bar{q})$ τ	Assets of marginal bank (\$ Billion)	50.138	[3.707]
	Cost of regulation (% of profit)	0.003	[0.012]

This table presents the results of the maximum likelihood estimation of the regulatory costs using the power-law distribution. The sample period is from 2018Q3 to 2019Q4. The estimation sample of panel A includes banks with assets between \$3 billion to \$40 billion. The estimation sample of panel B includes banks with assets above \$40 billion. Standard errors are obtained via the inverse of the Hessian matrix.

Then, we examine the effect of the 2018 regulatory relief on the indirect cost of regulation in column 3 of Table 5. We find that the regulatory relief alleviates the reduction in the number of banks. This result is consistent with the recovery in bank entries after 2018, as shown in Figure 7, panel A. Similarly, total output partially recovers but is still below the pre-Dodd-Frank Act equilibrium. Finally, the 2018 regulatory relief also partially reverses the distributional effects of the Dodd-Frank Act.

4.2 Alternative distribution assumption

In our baseline estimation, we assume that the undistorted assets follow a power-law distribution. This assumption is consistent with the equilibrium property of our model and the data. Nevertheless, we evaluate the sensitivity of our results to this assumption. To this end, we reestimate the regulatory costs assuming the undistorted assets follow a lognormal distribution.³⁴ Table 10 presents the results. We find that the estimated regulatory costs are similar using this alternative distribution assumption. Our estimates are insensitive to the distribution assumption because regulatory costs are mainly identified by the local abnormal densities.

4.3 Placebo tests

To further examine the robustness of our estimation, we conduct several placebo tests. First, we apply our maximum likelihood estimator to the pre-Dodd-Frank sample in panels A and B of Table 11. We set the standard deviation of the structural error σ to the value from our baseline estimations because it is difficult to identify this parameter without abnormal density created by regulations.³⁵ Nevertheless, the results are not sensitive to the

³⁴ The derivation of the maximum likelihood estimator under lognormal distribution can be found in Appendix A.6.

³⁵ Intuitively, this parameter measures the extent to which the abnormal densities are diffused around the threshold, as shown in Figure 5. Therefore, it is difficult to identify this parameter without abnormal density in the first place.

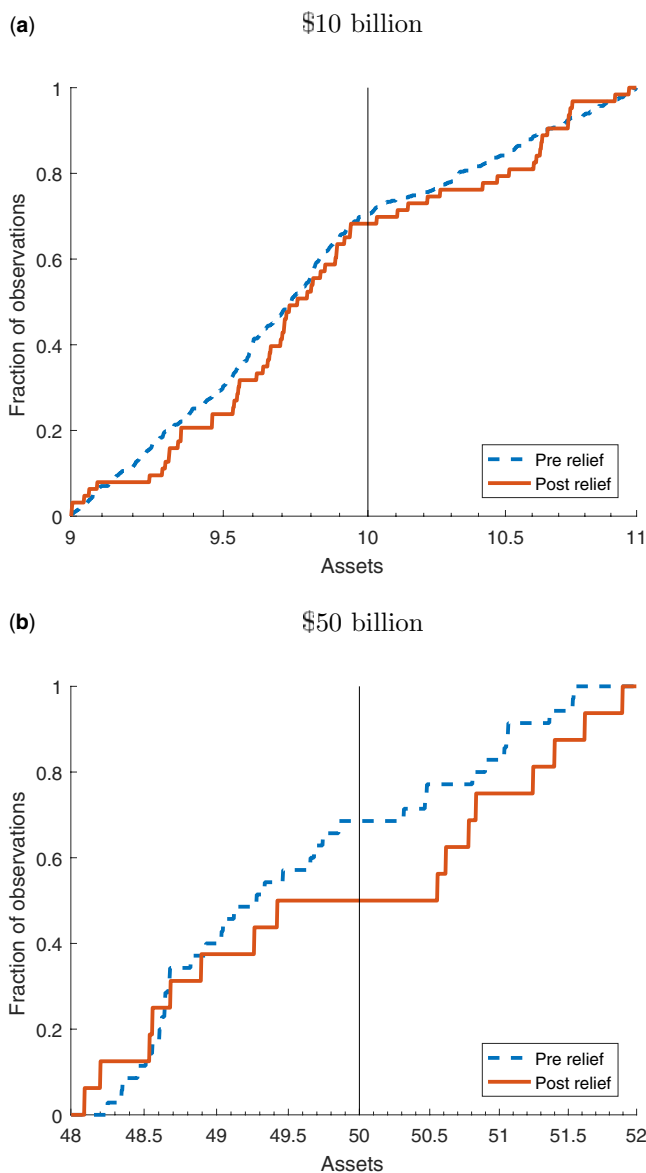


Figure 10

The size distribution of banks around regulatory thresholds before and after 2018 regulatory relief

This figure shows the empirical cumulative distribution of bank size conditional on the narrow intervals around the \$10 billion and \$50 billion regulatory thresholds. The blue dashed line and the red solid line represent the cumulative distribution functions before and after the regulatory relief in 2018Q2. Bank size is measured by quarter-end assets. The sample period is 2018Q3 to 2019Q4 for the postrelief period, and from 2010Q3 to 2018Q2 for the prerelief period.

Table 10
Maximum likelihood estimation of regulatory costs: Lognormal distribution

A. \$10 billion threshold			
		Estimated value	S.E.
μ_q	Average undistorted log(asset)	1.999	[0.007]
σ_q	Undistorted log(asset) std.	0.674	[0.005]
σ	Measurement error volatility (in %)	4.027	[0.378]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	10.888	[0.086]
τ	Cost of regulation (% of profit)	0.342	[0.062]
B. \$50 billion threshold			
		Estimated value	S.E.
μ_q	Average undistorted log(asset)	5.040	[0.022]
σ_q	Undistorted log(asset) std.	1.062	[0.016]
σ	Measurement error volatility (in %)	2.355	[0.533]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	52.552	[0.533]
τ	Cost of regulation (% of profit)	0.120	[0.048]

This table presents the results of the maximum likelihood estimation of the regulatory costs using the lognormal distribution. The sample period is from 2010Q3 to 2018Q2. The estimation sample of panel A includes banks with assets between \$3 billion to \$40 billion. The estimation sample of panel B includes banks with assets above \$40 billion. Standard errors are obtained via the inverse of the Hessian matrix.

value of this parameter. The results in panels A and B show that banks incur no additional regulatory cost when they cross the \$10 billion or \$50 billion thresholds before the Dodd-Frank Act. We also apply our maximum likelihood estimator to round numbers that are not regulatory thresholds, such as \$20 billion and \$40 billion. Our estimator correctly indicates null results, as shown in panels C and D.

4.4 Approximation error

Proposition 1 shows that the sufficient statistic formula (8) is exact when the semielasticity does not depend on bank size choice q . However, if the semielasticity is size dependent, then the sufficient statistic formula (8) leads to an approximation error.³⁶ We evaluate the magnitude of this approximation error using additional moments from the data. Specifically, the marginal buncher for the \$10 billion threshold has a size of \$11 billion according to Table 2. We calibrate the change in the inverse semielasticity, r_{qq} , to -0.2% using the data around the marginal buncher.³⁷ The approximation error at the \$10 billion threshold is 0.039%, which is small relative to the baseline estimate of the regulatory cost of 0.406%. This exercise suggests that the approximation error is economically small because the semielasticity does not seem to change substantially in the small bunching range.

³⁶ The approximation error is $\frac{1}{2(R-r(q;\bar{q}))}r_{qq}\left(q-\bar{q}\right)^2\exp\left(q-\bar{q}\right)$. See Appendix A.2 for derivation.

³⁷ We use a bin of \$1 billion centering around \$10 billion and \$11 billion in the pre-Dodd Frank sample to estimate the average profit margins at these two sizes, which are 2.3% and 2.1%, respectively. This implies that the change in the inverse semielasticity, r_{qq} , for the marginal buncher is -0.2% .

Table 11
Regulatory cost estimates: Placebo tests

<i>A. \$10 billion threshold, pre-Dodd-Frank</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.002]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	10.002	[0.239]
τ	Cost of regulation (% of profit)	0.005	[0.003]
<i>B. \$50 billion threshold, pre-Dodd-Frank</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.085	[0.003]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	51.221	[0.717]
τ	Cost of regulation (% of profit)	0.029	[0.034]
<i>C. \$20 billion threshold, post-Dodd-Frank</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.001]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	20.016	[0.618]
τ	Cost of regulation (% of profit)	0.004	[0.003]
<i>D. \$40 billion threshold, post-Dodd-Frank</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.085	[0.002]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	40.509	[1.546]
τ	Cost of regulation (% of profit)	0.008	[0.071]

This table presents the results of the maximum likelihood estimation of the regulatory costs using the power-law distribution. The sample periods of panels A and B are from 2001Q1 to 2007Q4. The sample periods of panels C and D are from 2010Q3 to 2017Q4. The estimation sample of panels A and C includes banks with assets between \$3 billion and \$40 billion. The estimation sample of panel B includes banks with assets above \$40 billion. The estimation sample of panel D includes banks with assets above \$30 billion. Standard errors are obtained via the inverse of the Hessian matrix.

4.5 Transition dynamics

Our baseline estimation assumes that the local size distribution around the threshold has already finished adjusting to the regulatory change one year after the Dodd-Frank Act. However, if there are convex adjustment costs, the bank size distribution may take more time to adjust to the regulatory change. As a result, the still-in-progress changes in the size distribution may bias our estimates downward. To address this concern, we first plot bank size distribution year by year in Figure OA.2. We can see that the bunching pattern starts to appear one year after the introduction of the Dodd-Frank Act. This bunching pattern suggests that bank size adjustment was quite fast, possibly because banks had liquid assets, such as marketable securities, and short-term liabilities, such as wholesale funding, which allowed the adjustment to happen quickly. It is worth noting that our estimation does not require the entire bank size distribution to be stationary after the regulatory change. It only requires that the normalized local distribution around the regulatory threshold has finished adjusting to the Dodd-Frank Act before the start of the estimation sample.

To further address the concern about the still-in-progress changes in the size distribution, in Table 12, we drop the first one, two, and three years after the

Table 12
Maximum likelihood estimation of regulatory costs at \$10 billion: Transition dynamics

<i>A. Drop first year</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.001]
σ	Measurement error volatility (in %)	4.236	[0.411]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	11.000	[0.090]
τ	Cost of regulation (% of profit)	0.426	[0.071]
<i>B. Drop first two years</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.002]
σ	Measurement error volatility (in %)	4.255	[0.453]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	11.040	[0.097]
τ	Cost of regulation (% of profit)	0.458	[0.079]
<i>C. Drop first three years</i>			
		Estimated value	S.E.
β	Exponent of the power-law distribution	1.112	[0.002]
σ	Measurement error volatility (in %)	4.027	[0.543]
$\exp(\bar{q})$	Assets of marginal bank (\$ Billion)	10.998	[0.110]
τ	Cost of regulation (% of profit)	0.425	[0.087]

This table presents the results of the maximum likelihood estimation of the regulatory costs using the power-law distribution. The sample period is 2011Q3–2018Q2, 2012Q3–2018Q2, and 2013Q3–2018Q2 for panels A, B, and C, respectively. The estimation sample includes banks with assets between \$3 billion and \$40 billion. Standard errors are obtained via the inverse of the Hessian matrix.

Dodd-Frank Act and rerun the estimation. We find that the estimated regulatory costs are quite similar to the baseline estimates when the whole post-Dodd-Frank sample is included. This result again suggests that bank size adjustment is relatively fast.

In addition to the lagged response, a related issue is anticipatory responses. If banks face convex adjustment costs and cannot adjust their size quickly, they may change their size before the regulation comes into effect, which may bias the estimates if the preregulation data are used to estimate the counterfactual distribution. To address the possible anticipatory responses, we elect to use only the post-Dodd-Frank sample in our baseline estimation. Nevertheless, the results are robust if the pre-Dodd-Frank sample is included, as shown in Table OA.3.

4.6 Alternative measurement of bank regulatory assets

As discussed in Section 1.3, different provisions of the Dodd-Frank Act use slightly different methodologies to measure size to determine banks' regulatory status. Without a uniform way to measure the size for all the provisions, we use the simplest approach of quarter-end assets to measure bank size. In this section, we evaluate the sensitivity of the estimated regulatory cost to three alternative measures of bank assets: (1) the average of the last four quarters, (2) the end of the last calendar year, (3) the minimum of the last four quarters in Table OA.6. We find very similar estimates of the regulatory costs as our baseline results. The reason is twofold: (1) these measures only differ slightly

from each other in the data, and (2) we explicitly account for measurement error in the maximum likelihood estimation. Indeed, the differences between alternative measures mainly show up in the estimated standard deviation of the measurement error, σ , rather than the regulatory costs.

5. Conclusion

In this paper, we propose a revealed preference approach to estimate regulatory costs borne by financial institutions. By focusing on banks' actions rather than their self-reported estimates, our approach circumvents the information obstacle faced by regulators. We use our approach to estimate the costs of the Dodd-Frank Act. We find that the regulation costs triggered at the \$10 billion threshold are equivalent to a 0.41% tax on banks' average annual profits. The regulatory costs triggered at the \$50 billion threshold are equivalent to a 0.11% tax. In total, the regulatory costs introduced by the Dodd-Frank Act for a \$50 billion bank amount to \$4.16 million per year. We also examine the impacts of the regulation on bank franchise value, entry and exits, and the output of bank-dependent firms. Although our estimated regulatory costs are substantial, they are significantly lower than those claimed by banks. Our estimates of regulatory costs can be used in conjunction with estimates of regulatory benefits in CBA. Overall, this paper shows that regulators can extract valuable information by analyzing financial institutions' endogenous response to regulations. This revealed preference approach can be used in many settings in which the regulatory changes are big and highly anticipated.

A. Appendix

A.1 Institutional Background of the CBA of Financial Regulation

The major statutes that apply to financial regulators include (1) the Paperwork Reduction Act (PRA), which requires regulators to justify the collection of information from the public to minimize its burden and maximize the utility of information collected;³⁸ (2) the Regulatory Flexibility Act (RFA), which requires regulators to assess and consider alternatives to the burden of regulation on small entities;³⁹ and (3) the Congressional Review Act (CRA), which requires regulators to submit proposed rules—along with any cost-benefit analysis the agencies have conducted—to Congress and the Government Accountability Office (GAO).⁴⁰

The cost-benefit analysis often provides grounds for judicial review and Congressional oversight of regulatory actions. For instance, in the 2002 SEC mutual fund governance reform, the regulator proposed to increase the required share of independent directors from 50% to 75% for mutual funds. Following the SEC's proposal, the Chamber of Commerce, a business-oriented interest group, sued the SEC for failure to quantify the costs of the rule. In its defense, the SEC argued that staffing would be discretionary, and the SEC had no basis for knowing how many chairs would hire staff or how many staff each chair would hire. However, the court rejected this argument

³⁸ Pub. L. No. 104–13, 109 Stat. 163 (1995) (codified at 44 U.S.C. §§3501–3520).

³⁹ Pub. L. No. 96–354, 94 Stat. 1164 (1980) (codified at 5 U.S.C. §§601–612).

⁴⁰ Pub. L. No. 104–121, 110 Stat. 847 (1996) (codified as amended in 5 U.S.C. §801 et seq.).

and maintained that “estimating the costs for an individual fund is pertinent to an assessment of the requirement.”⁴¹ In the end, the rule was struck down by the court.

A.2 Derivation of Regulatory Costs

Provided the different regulatory thresholds are far enough from each other, we can derive the regulatory cost τ_i in the same manner for all i . Therefore, we drop the subscript i in the following derivation. We know from the profit indifference condition of the marginal bank in Equation (7) that τ can be written as follows:

$$1 - \tau = \frac{(R - r(\underline{q}|\bar{z}))\exp(\underline{q})}{(R - r(\bar{q}|\bar{z}))\exp(\bar{q})}. \quad (\text{A1})$$

We conduct a Taylor expansion for $r(\underline{q}|\bar{z})$ at the optimal size, $\bar{q}$

$$r(\underline{q}|\bar{z}) = r(\bar{q}|\bar{z}) + r_q(\bar{q}|\bar{z})(\underline{q} - \bar{q}) + O((\underline{q} - \bar{q})^2). \quad (\text{A2})$$

The numerator of Equation (A1) becomes

$$\begin{aligned} (R - r(\underline{q}|\bar{z}))\exp(\underline{q}) &= \left\{ R - r(\bar{q}|\bar{z}) - r_q(\bar{q}|\bar{z})(\underline{q} - \bar{q}) - O((\underline{q} - \bar{q})^2) \right\} \exp(\underline{q}) \\ &= \left\{ (R - r(\bar{q}|\bar{z}))(1 - \underline{q} + \bar{q}) - O((\underline{q} - \bar{q})^2) \right\} \exp(\underline{q}), \end{aligned} \quad (\text{A3})$$

where the second equality uses the result of Equation (2) that the profit margin at the optimal size equals the inverse semielasticity $R - r(\bar{q}|\bar{z}) = r_q(\bar{q}|\bar{z})$. Plugging Equation (A3) into Equation (A1), we get

$$1 - \tau = (1 - \underline{q} + \bar{q})\exp(\underline{q} - \bar{q}) - O((\underline{q} - \bar{q})^2 \exp(\underline{q} - \bar{q})). \quad (\text{A4})$$

Thus, when $\underline{q}$ is close to $\bar{q}$, we get

$$1 - \tau \simeq (\bar{q} - \underline{q} + 1)\exp(\underline{q} - \bar{q}). \quad (\text{A5})$$

When the inverse semielasticity of funding supply does not depend on the size, that is, $r_{qq} = 0$, the higher-order term $O((\underline{q} - \bar{q})^2 \exp(\underline{q} - \bar{q})) = \frac{1}{2(R - r(\bar{q}|\bar{z}))} r_{qq}(\bar{q}|\bar{z}) (\underline{q} - \bar{q})^2 \exp(\underline{q} - \bar{q})$ in the Taylor expansion disappears and the approximation becomes exact.

A.3 Heterogeneity in Banks' Profit Margins

This appendix considers the effect of heterogeneous demand semielasticity on the sufficient statistic formula of regulatory cost, Equation (8). Suppose the funding supply function is $r(q|z, \theta) = \frac{1}{\theta}(q - z)$ and the semielasticity, θ , follows a distribution $x(\theta)$ so that it is heterogeneous across banks. Consider the bunching decision for a subsample of banks which has semielasticity θ :

$$\max_q \pi(q|z, \theta) = \max_q \left(R - \frac{1}{\theta}(q - z) \right) \exp(q) \cdot (1 - \tau \mathbb{1}_{q \geq \underline{q}}). \quad (\text{A6})$$

The indifference condition between bunching and not bunching of the marginal bank within this subset is given by

$$1 - \tau = \frac{\left(R - \frac{1}{\theta}(\underline{q} - \bar{z}_\theta) \right) \exp(\underline{q})}{\left(R - \frac{1}{\theta}(\bar{q} - \bar{z}_\theta) \right) \exp(\bar{q})}, \quad (\text{A7})$$

⁴¹ Chamber of Commerce v. SEC, 412 F.3d 133 (D.C. Cir. 2005).

where the optimal size when the bank does not bunch is given by

$$\bar{q}_\theta = \theta R + \bar{z}_\theta - 1. \quad (\text{A8})$$

The subscript θ indicates that the log asset size and the productivity may potentially depend on θ . Using the above optimal size equation to replace $\bar{z}_\theta$ with $\bar{q}_\theta$ in the first term in the numerator of Equation (A7), we get

$$\begin{aligned} & R - \frac{1}{\theta}(\underline{q} - \bar{z}_\theta) \\ &= R - \frac{1}{\theta}(\underline{q} - \bar{q}_\theta + \theta R - 1) \\ &= \frac{1}{\theta}(\bar{q}_\theta - \underline{q} + 1) \end{aligned} \quad (\text{A9})$$

The first term in the denominator of Equation (A7) is the undistorted profit margin, which equals the reciprocal of the semielasticity:

$$R - \frac{1}{\theta}(\bar{q}_\theta - \bar{z}_\theta) = \frac{1}{\theta}. \quad (\text{A10})$$

We can plug the above two equations into Equation (A7), and we get

$$1 - \tau = \frac{\frac{1}{\theta}(\bar{q}_\theta - \underline{q} + 1) \exp(\underline{q})}{\frac{1}{\theta} \exp(\bar{q}_\theta)}, \quad (\text{A11})$$

The semielasticity cancels out and we get

$$\tau = 1 - (\bar{q}_\theta - \underline{q} + 1) \exp(\underline{q} - \bar{q}_\theta). \quad (\text{A12})$$

Note that the size of the marginal bank only depends on τ , but not on θ . Therefore, we can drop the subscript θ , and we get the same sufficient statistic formula as Equation (8):

$$\tau = 1 - (\bar{q} - \underline{q} + 1) \exp(\underline{q} - \bar{q}). \quad (\text{A13})$$

In other words, the size of the marginal bank in the subsample of banks with semielasticity θ is the same as that in a different subsample of banks with a different semielasticity θ' . The heterogeneity in θ is absorbed by z . The above derivation shows that the mapping between the proportional regulatory tax, τ , and the marginal bunching banks' size, $\bar{q}$, does not depend on the value of semielasticity θ . As a result, the heterogeneous semielasticity does not affect the estimate of the proportional regulatory tax. However, the heterogeneous semielasticity does affect the dollar value of the regulatory costs, $\tau \frac{1}{\theta} \exp(\underline{q})$. Intuitively, for a given percentage of assets that banks are willing to give up, a lower semielasticity implies that banks forego more profits. Thus, the implied dollar value of regulatory cost is higher. This heterogeneity can be accounted for by multiplying the proportional tax rate with bank-specific expected profits.

A.4 Derivation of the Equilibrium Distribution

Guess $g(z) = c_z \exp(z)^{-\beta}$ and plug it to the Kolmogorov forward Equation (14):

$$0 = -\frac{\partial}{\partial z} [\mu_z c_z \exp(z)^{-\beta}] + \frac{1}{2} \frac{\partial^2}{\partial z^2} [\sigma_z^2 c_z \exp(z)^{-\beta}] - \lambda c_z \exp(z)^{-\beta}, \quad (\text{A14})$$

for $z \neq z_n$. This leads to a quadratic equation

$$0 = -\mu_z \beta + \frac{1}{2} \sigma_z^2 \beta (\beta + 1) - \lambda. \quad (\text{A15})$$

This quadratic equation always has two real roots $\beta_+ > 0$ and $\beta_- < 0$ as long as $\lambda > 0$. The general solution to the Kolmogorov forward equation is

$$g(z) = c_- \exp(z)^{-\beta_-} + c_+ \exp(z)^{-\beta_+}. \quad (\text{A16})$$

$g(z)$ is integrable, which implies that $c_- = 0$ for $z > z_n$ and $c_+ = 0$ for $z < z_n$ (Moll 2018). Therefore, the equilibrium distribution of the productivity is a double power-law distribution.

$$g(z) = \begin{cases} c_z \exp(z - z_n)^{-\beta_-} & z < z_n \\ c_z \exp(z - z_n)^{-\beta_+} & z > z_n, \end{cases} \quad (\text{A17})$$

where z_n is the productivity of the new entrants.

Assuming the semielasticity of funding supply is constant, the undistorted log asset is a linear function of productivity according to Equation (3). The equilibrium distribution of bank size is also a double power-law distribution in the absence of regulatory distortion.

A.5 Derivation of the Maximum Likelihood Estimator: Power Law

Define h and H as the probability density function and cumulative distribution function of the undistorted bank log assets q . Using banks' optimal size choice described in Equation (4), the theoretical density function of banks' log assets with one regulatory threshold is given by the following distribution function:

$$q \sim \begin{cases} h(q) & q \in (-\infty, \underline{q}) \\ H(\bar{q}) - H(\underline{q}) & q = \underline{q} \\ 0 & q \in (\underline{q}, \bar{q}] \\ h(q) & q \in (\bar{q}, \infty). \end{cases} \quad (\text{A18})$$

Empirical bank size a is observed with a structural error $u = \sigma v$ with $v \sim N(0, 1)$. Denote $\mathbb{P}(x \leq a|v)$ as the probability of observing log assets x below a given value a for a given realization of the structural error v . We have

$$\mathbb{P}(x \leq a|v) = \mathbb{P}(q + \sigma v \leq a|v) = \mathbb{P}(q \leq a - \sigma v|v). \quad (\text{A19})$$

The above equation shows that the probability of observing size x below a given value a for a given realization of the structural error σv is the same as the probability that the theoretical size q is below $a - \sigma v$.

Using equations (A18) and (A19), the conditional cumulative distribution of the observed size a is given by

$$\mathbb{P}(x \leq a|v) = \begin{cases} H(a - \sigma v) & a - \sigma v \in (-\infty, \underline{q}) \\ H(\bar{q}) & a - \sigma v \in [\underline{q}, \bar{q}] \\ H(a - \sigma v) & a - \sigma v \in (\bar{q}, \infty). \end{cases} \quad (\text{A20})$$

We can change the conditions such that they are written with respect to the structural error, v :

$$\mathbb{P}(x \leq a|v) = \begin{cases} H(a - \sigma v) & v \in (\frac{1}{\sigma}(a - \underline{q}), \infty) \\ H(\bar{q}) & v \in [\frac{1}{\sigma}(a - \bar{q}), \frac{1}{\sigma}(a - \underline{q})] \\ H(a - \sigma v) & v \in (-\infty, \frac{1}{\sigma}(a - \bar{q})). \end{cases} \quad (\text{A21})$$

The unconditional cumulative distribution function of observed size a can be written as the convolution of the conditional CDF of q and the distribution of the structural error v :

$$\begin{aligned}\mathbb{P}(x \leq a) &= \mathbb{E}[\mathbb{P}(x \leq a | v)] \\ &= \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} H(a-\sigma v) \phi(v) dv \\ &\quad + H(\bar{q}) \left[\Phi\left(\frac{1}{\sigma}(a-\underline{q})\right) - \Phi\left(\frac{1}{\sigma}(a-\bar{q})\right) \right] \\ &\quad + \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} H(a-\sigma v) \phi(v) dv.\end{aligned}\tag{A22}$$

We take the derivative of Equation (A22) to get the probability density function of the observed size a :

$$\begin{aligned}f(a) &= \frac{d}{da} \mathbb{E}[\mathbb{P}(x \leq a | v)] \\ &= \frac{d}{da} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} H(a-\sigma v) \phi(v) dv \\ &\quad + \frac{d}{da} H(\bar{q}) \left[\Phi\left(\frac{1}{\sigma}(a-\underline{q})\right) - \Phi\left(\frac{1}{\sigma}(a-\bar{q})\right) \right] \\ &\quad + \frac{d}{da} \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} H(a-\sigma v) \phi(v) dv.\end{aligned}\tag{A23}$$

The second term in Equation (A23) is straightforward to compute

$$\begin{aligned}&\frac{d}{da} H(\bar{q}) \left[\Phi\left(\frac{1}{\sigma}(a-\underline{q})\right) - \Phi\left(\frac{1}{\sigma}(a-\bar{q})\right) \right] \\ &= \frac{1}{\sigma} H(\bar{q}) \left[\phi\left(\frac{1}{\sigma}(a-\underline{q})\right) - \phi\left(\frac{1}{\sigma}(a-\bar{q})\right) \right].\end{aligned}$$

For the first and third terms in Equation (A23), we use the Leibniz formula:

$$\frac{d}{da} \int_{t_1(a)}^{t_2(a)} f(a, v) dv = f(a, t_2(a)) t_2'(a) - f(a, t_1(a)) t_1'(a) + \int_{t_1(a)}^{t_2(a)} f_a(a, v) dv.$$

Using the Leibniz formula, the first term in Equation (A23) becomes

$$\frac{d}{da} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} H(a-\sigma v) \phi(v) dv = -\frac{1}{\sigma} H(\underline{q}) \phi\left(\frac{1}{\sigma}(a-\underline{q})\right) + \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} h(a-\sigma v) \phi(v) dv,$$

given that $\frac{1}{\sigma} H(-\infty) \cdot \phi(\infty) = 0$.

The third term in Equation (A23) becomes

$$\frac{d}{da} \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} H(a-\sigma v) \phi(v) dv = \frac{1}{\sigma} H(\bar{q}) \phi\left(\frac{1}{\sigma}(a-\bar{q})\right) + \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} h(a-\sigma v) \phi(v) dv,$$

given that $\frac{1}{\sigma} H(\infty) \cdot \phi(\infty) = 0$. In summary, the probability density of the observed bank size is given by

$$f(a) = -\frac{1}{\sigma} H(\underline{q}) \phi\left(\frac{1}{\sigma}(a-\underline{q})\right) + \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} h(a-\sigma v) \phi(v) dv$$

$$\begin{aligned}
 & + \frac{1}{\sigma} H(\bar{q}) \left[\phi \left(\frac{1}{\sigma} (a - \underline{q}) \right) - \phi \left(\frac{1}{\sigma} (a - \bar{q}) \right) \right] \\
 & + \frac{1}{\sigma} H(\bar{q}) \phi \left(\frac{1}{\sigma} (a - \bar{q}) \right) + \int_{-\infty}^{\frac{1}{\sigma} (a - \bar{q})} h(a - \sigma v) \phi(v) dv.
 \end{aligned}$$

By simplifying the terms, we have

$$\begin{aligned}
 f(a) = & \underbrace{\frac{1}{\sigma} \left[\phi \left(\frac{1}{\sigma} (a - \underline{q}) \right) \right] (H(\bar{q}) - H(\underline{q}))}_{\text{Term 1}} \\
 & + \underbrace{\int_{\frac{1}{\sigma} (a - \underline{q})}^{\infty} h(a - \sigma v) \phi(v) dv}_{\text{Term 2}} \\
 & + \underbrace{\int_{-\infty}^{\frac{1}{\sigma} (a - \bar{q})} h(a - \sigma v) \phi(v) dv}_{\text{Term 3}}.
 \end{aligned} \tag{A24}$$

We now assume the undistorted bank assets follow a power-law distribution, $\exp(q) \sim c \exp(q)^{-\beta}$ over the strictly positive support $[q_{\min}, \infty]$, where $q_{\min}$ corresponds to the minimum theoretical log asset of a bank that could be observed. Given that we defined q as log assets of a bank (see Section 2), the probability density function of q is

$$\begin{aligned}
 h(q) &= \frac{d}{dq} \mathbb{P}(\exp(x) \leq \exp(q)) \\
 &= \frac{d}{dq} \left(\frac{c}{1 - \beta} e^{(1 - \beta)q} \right) \\
 &= c \exp((1 - \beta)q),
 \end{aligned} \tag{A25}$$

for $q \geq q_{\min}$, and $h(q) = 0$ for $q < q_{\min}$. The scaling parameter c is constrained to be equal to $(\beta - 1) / \exp((1 - \beta)q_{\min})$ such that the distribution integrates to one.

Plugging Equation (A25) into Equation (A24), we can derive the first term of Equation (A24) as

$$\text{Term 1} = \frac{1}{\sigma} \frac{c}{1 - \beta} \left[\phi \left(\frac{1}{\sigma} (a - \underline{q}) \right) \right] \left(\exp((1 - \beta)\bar{q}) - \exp((1 - \beta)\underline{q}) \right).$$

The second term of Equation (A24) is

Term 2

$$\begin{aligned}
 &= \int_{\frac{1}{\sigma} (a - \underline{q})}^{\infty} \mathbf{1}_{\{a - \sigma v \geq q_{\min}\}} c \exp((1 - \beta)(a - \sigma v)) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}v^2\right) dv \\
 &= c \exp\left((1 - \beta)a + \frac{1}{2}(\beta - 1)^2 \sigma^2\right) \int_{\frac{1}{\sigma} (a - \underline{q})}^{\infty} \mathbf{1}_{\{v \leq \frac{1}{\sigma} (a - q_{\min})\}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(v - (\beta - 1)\sigma)^2\right) dv \\
 &= c \exp\left((1 - \beta)a + \frac{1}{2}(\beta - 1)^2 \sigma^2\right) \cdot \\
 &\quad \left(\Phi\left(\frac{1}{\sigma}(a - q_{\min}) - (\beta - 1)\sigma\right) - \Phi\left(\frac{1}{\sigma}(a - \underline{q}) - (\beta - 1)\sigma\right) \right).
 \end{aligned}$$

The third term of Equation (A24) is

$$\begin{aligned} \text{Term 3} &= \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} \mathbf{1}_{\{a-\sigma v \geq q_{\min}\}} c \exp((1-\beta)(a-\sigma v)) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}v^2\right) dv \\ &= c \exp\left((1-\beta)a + \frac{1}{2}(\beta-1)^2\sigma^2\right) \Phi\left(\frac{1}{\sigma}(a-\bar{q}) - (\beta-1)\sigma\right). \end{aligned}$$

To summarize, the probability density of the observed bank size is given by

$$\begin{aligned} f(a) &= \frac{1}{\sigma} \frac{c}{1-\beta} \left[\phi\left(\frac{1}{\sigma}(a-\underline{q})\right) \right] \left(\exp((1-\beta)\bar{q}) - \exp((1-\beta)\underline{q}) \right) \\ &\quad + c \exp\left((1-\beta)a + \frac{1}{2}(\beta-1)^2\sigma^2\right) \\ &\quad \left(\Phi\left(\frac{1}{\sigma}(a-q_{\min}) - (\beta-1)\sigma\right) - \Phi\left(\frac{1}{\sigma}(a-\underline{q}) - (\beta-1)\sigma\right) + \Phi\left(\frac{1}{\sigma}(a-\bar{q}) - (\beta-1)\sigma\right) \right). \end{aligned} \quad (\text{A26})$$

Taking the logarithm of Equation (A26), we can get the log-likelihood function for the MLE. Note that when $\tau=0$, we have $\bar{q}=\underline{q}$. The first term of Equation (A26) equals zero, and this equation simplifies to

$$f(a) = c \exp\left((1-\beta)a + \frac{1}{2}(\beta-1)^2\sigma^2\right) \Phi\left(\frac{1}{\sigma}(a-q_{\min}) - (\beta-1)\sigma\right).$$

A.6 Derivation of the Maximum Likelihood Estimator: Lognormal

The derivation of the maximum likelihood estimator under the lognormal distribution is the same as the power-law distribution until Equation (A25), where the undistorted asset size follows a lognormal distribution instead. As a result, the undistorted log assets q follows a normal distribution with probability density function:

$$h(q) = \frac{1}{\sigma_q} \phi\left(\frac{q - \mu_q}{\sigma_q}\right), \quad (\text{A27})$$

where μ_q denotes the mean undistorted log assets, and σ_q denotes the standard deviation. Plugging Equation (A27) into Equation (A24), we can derive the first term of Equation (A24) as

$$\text{Term 1} = \frac{1}{\sigma} \left[\phi\left(\frac{1}{\sigma}(a-\underline{q})\right) \right] \left(\Phi\left(\frac{\bar{q} - \mu_q}{\sigma_q}\right) - \Phi\left(\frac{\underline{q} - \mu_q}{\sigma_q}\right) \right).$$

The second term of Equation (A24) is

$$\begin{aligned} \text{Term 2} &= \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \frac{1}{\sigma_q} \phi\left(\frac{a-\sigma v - \mu_q}{\sigma_q}\right) \phi(v) dv \\ &= \frac{1}{2\pi} \frac{1}{\sigma_q} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \exp\left(-\frac{1}{2}\left(\frac{a-\sigma v - \mu_q}{\sigma_q}\right)^2 - \frac{1}{2}v^2\right) dv \\ &= \frac{1}{2\pi} \frac{1}{\sigma_q} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \exp\left(-\frac{1}{2} \frac{(\sigma^2 + \sigma_q^2)v^2 - 2(a - \mu_q)\sigma v + (a - \mu_q)^2}{\sigma_q^2}\right) dv \\ &= \frac{1}{2\pi} \frac{1}{\sigma_q} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \exp\left(-\frac{1}{2} \frac{(\sigma^2 + \sigma_q^2)\left(v - \frac{a - \mu_q}{\sigma^2 + \sigma_q^2}\sigma\right)^2 - \frac{(a - \mu_q)^2}{\sigma^2 + \sigma_q^2}\sigma^2 + (a - \mu_q)^2}{\sigma_q^2}\right) dv \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sigma_q} \exp \left(\frac{\left(\frac{a-\mu_q}{\sigma^2+\sigma_q^2} \right)^2 \sigma^2 - (a-\mu_q)^2}{2\sigma_q^2} \right) \frac{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}}{\sqrt{2\pi}} \\
 &\quad \cdot \frac{1}{\sqrt{2\pi \frac{\sigma_q^2}{(\sigma^2+\sigma_q^2)}}} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \exp \left(-\frac{1}{2} \left(\frac{v - \frac{a-\mu_q}{\sigma^2+\sigma_q^2} \sigma}{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \right)^2 \right) dv \\
 &= \frac{1}{\sigma_q} \frac{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} \frac{(a-\mu_q)^2}{\sigma^2+\sigma_q^2} \right) \\
 &\quad \cdot \frac{1}{\sqrt{2\pi \frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \int_{\frac{1}{\sigma}(a-\underline{q})}^{\infty} \exp \left(-\frac{\sigma^2+\sigma_q^2}{2\sigma_q^2} \left(v - \sigma \frac{a-\mu_q}{\sigma^2+\sigma_q^2} \right)^2 \right) dv \\
 &= \frac{1}{\sqrt{2\pi (\sigma^2+\sigma_q^2)}} \exp \left(-\frac{1}{2} \frac{(a-\mu_q)^2}{\sigma^2+\sigma_q^2} \right) \left\{ 1 - \Phi \left(\frac{\frac{1}{\sigma}(a-\underline{q}) - \sigma \frac{a-\mu_q}{\sigma^2+\sigma_q^2}}{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \right) \right\}.
 \end{aligned}$$

The third term of Equation (A24) is computed similarly

$$\begin{aligned}
 \text{Term 3} &= \int_{-\infty}^{\frac{1}{\sigma}(a-\bar{q})} \frac{1}{\sigma_q} \phi \left(\frac{a-\sigma v - \mu_q}{\sigma_q} \right) \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} v^2 \right) dv \\
 &= \frac{1}{\sqrt{2\pi (\sigma^2+\sigma_q^2)}} \exp \left(-\frac{1}{2} \frac{(a-\mu_q)^2}{\sigma^2+\sigma_q^2} \right) \Phi \left(\frac{\frac{1}{\sigma}(a-\bar{q}) - \sigma \frac{a-\mu_q}{\sigma^2+\sigma_q^2}}{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \right).
 \end{aligned}$$

In summary, the probability density of the observed bank log assets is given by

$$\begin{aligned}
 f(a) &= \frac{1}{\sigma} \phi \left(\frac{1}{\sigma} (a-\underline{q}) \right) \left(\Phi \left(\frac{\bar{q}-\mu_q}{\sigma_q} \right) - \Phi \left(\frac{\underline{q}-\mu_q}{\sigma_q} \right) \right) \\
 &\quad + \frac{1}{\sqrt{2\pi (\sigma^2+\sigma_q^2)}} \exp \left(-\frac{1}{2} \frac{(a-\mu_q)^2}{\sigma^2+\sigma_q^2} \right) \left\{ 1 - \Phi \left(\frac{\frac{1}{\sigma}(a-\underline{q}) - \sigma \frac{a-\mu_q}{\sigma^2+\sigma_q^2}}{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \right) \right\} \\
 &\quad + \frac{1}{\sqrt{2\pi (\sigma^2+\sigma_q^2)}} \exp \left(-\frac{1}{2} \frac{(a-\mu_q)^2}{\sigma^2+\sigma_q^2} \right) \Phi \left(\frac{\frac{1}{\sigma}(a-\bar{q}) - \sigma \frac{a-\mu_q}{\sigma^2+\sigma_q^2}}{\sqrt{\frac{\sigma_q^2}{\sigma^2+\sigma_q^2}}} \right).
 \end{aligned} \tag{A28}$$

Taking the logarithm of Equation (A28), we can get the log-likelihood function for the MLE. Note that when $\tau = 0$, we have $\bar{q} = q$. The first term of Equation (A28) equals zero, and this equation simplifies to

$$f(a) = \frac{1}{\sqrt{2\pi(\sigma^2 + \sigma_q^2)}} \exp\left(-\frac{1}{2} \frac{(a - \mu_q)^2}{\sigma^2 + \sigma_q^2}\right).$$

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